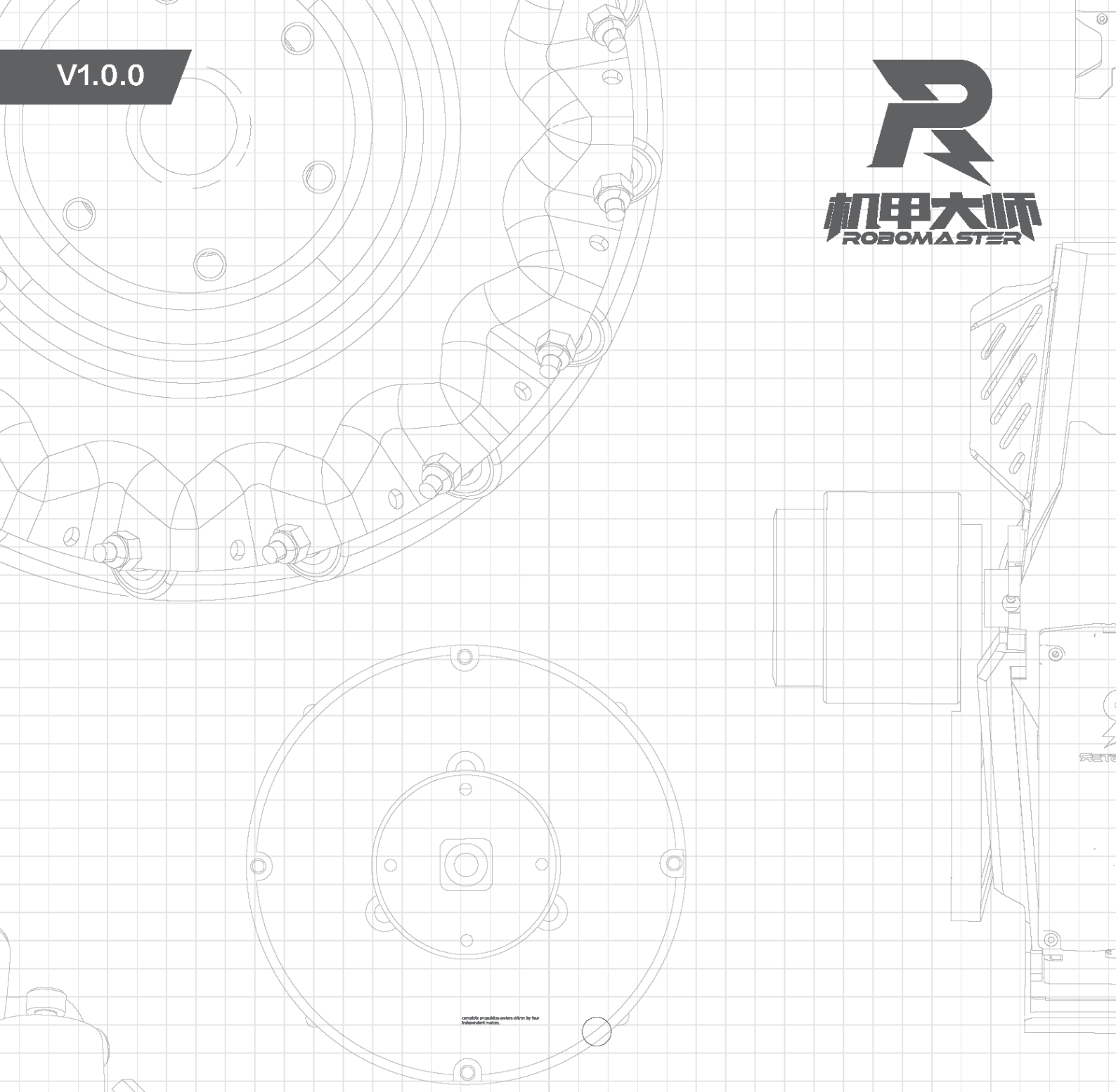
V1.0.0



ROBOMASTER 2026

University Series

# Communication Protocol

Prepared by the RoboMaster Organizing Committee

Released in December, 2025

# Release Notes

| Date | Version | Revision History |
|---|---|---|
| 2025.12.03 | V1.0 | First release |

# Foreword

The scope of this communication protocol for RMUC and RMUL competitions is as follows:

- For entries shared by both RMUC and RMUL (e.g., robot types, mechanics, or battlefield components), the provisions of this protocol apply to both events;
- For entries that apply exclusively to one event shall, by default, not apply to the other.

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# 1. Serial communication protocol

## 1.1 Serial communication protocol format

The robots communicate via a serial port protocol with the following configuration: for standard data link, the baud rate is 115200; for VTM link, the baud rate is 921600; 8 data bits, 1 stop bit, no hardware flow control, and no parity bit.

Table 1-1 Communication protocol format

| frame_header | cmd_id | data | frame_tail |
|---|---|---|---|
| 5-byte | 2-byte | n-byte | 2-byte, CRC16, package-level check |

Table 1-2 frame_header format

| SOF | data length | seq | CRC8 |
|---|---|---|---|
| 1 byte | 2-byte | 1 byte | 1 byte |

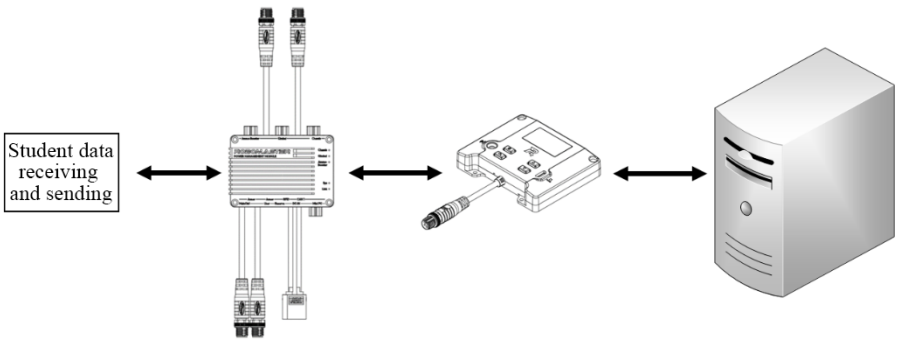
Table 1-3 Detailed definition of frame header

| Field | Offset location | Size (bytes) | Detailed description |
|---|---|---|---|
| SOF | 0 | 1 | Start byte of the data frame, fixed value is 0xA5 |
| data length | 1 | 2 | Length of data in the data frame |
| seq | 3 | 1 | Packet Sequence Number |
| CRC8 | 4 | 1 | Frame header CRC8 verification |

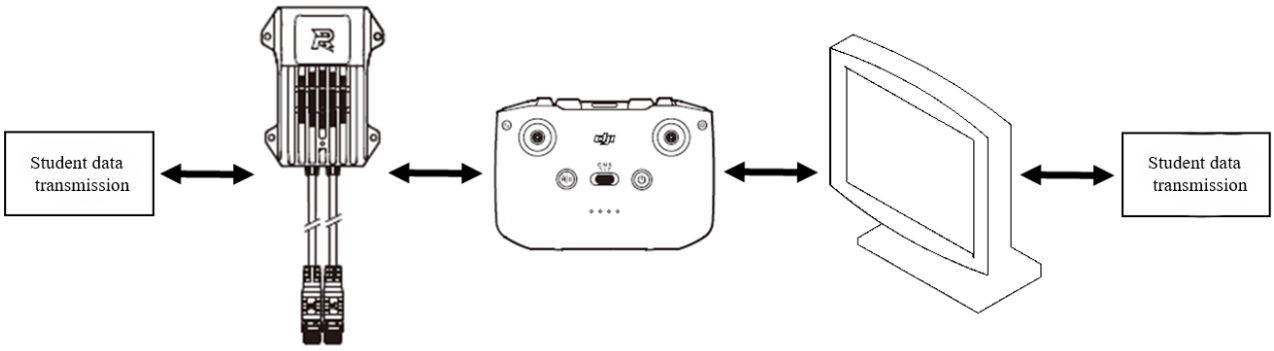
There are two types of Referee System serial data links: standard data link and VTM link.

- Standard data link relays data between the Referee System Server and the Main Control Module, with data sent and received via the Power Management Module's User serial port. The graph is shown below:




- The VTM link relays data between the Referee System Player's Client and the VTM Module, receiving data from the serial port of the VTM Module (Transmitter). See the following graph for details




- Under normal operating conditions, Referee System data latency is approximately 130 ms, with a packet loss rate of less than 1%.
- When the battlefield network environment is poor, Referee System data latency is approximately 200 ms, and the packet loss rate is around 3%.
- Measurement data may contain errors and are for reference only.

# 1.2 Instructions for command code ID and standard data link

Table 1-4 Overview of command code IDs

| Command code | Data segment length | Description | Sender/Receiver | Data Link |
|---|---|---|---|---|
| 0x0001 | 11 | Competition status data, sent at a fixed frequency of 1 Hz. | Server → All robots | Data Link |
| 0x0002 | 1 | Match result data is sent when end of round is Triggered. | Server → All robots | Data Link |
| 0x0003 | 16 | Robots' HP data, sent at a fixed rate of 3 Hz. | Server → All robots | Data Link |
| 0x0101 | 4 | Match event data, transmitted at a fixed frequency of 1 Hz. | Server → All own side robots | Data Link |
| 0x0104 | 3 | Referee warning data, sent upon trigger by penalty or forfeiture for own side, and transmitted at a frequency of 1 Hz at other times. | Server → All robots on the penalized side | Data Link |
| 0x0105 | 3 | Dart launch-related data, sent at a fixed frequency of 1 Hz. | Server → All own side robots | Data Link |
| 0x0201 | 13 | Robots performance system data, sent at a fixed frequency of 10 Hz. | Main Control Module → respective robot | Data Link |
| 0x0202 | 14 | Real-time chassis buffer energy and barrel heat data, sent at a fixed frequency of 10 Hz. | Main Control Module → respective robot | Data Link |
| 0x0203 | 16 | Robots location data, transmitted at a fixed frequency of 1 Hz | Main Control Module → respective robot | Data Link |

| Command code | Data segment length | Description | Sender/Receiver | Data Link |
|---|---|---|---|---|
| 0x0204 | 8 | Robots' gain and chassis energy data, sent at a fixed frequency of 3 Hz | Server → respective robot | Data Link |
| 0x0206 | 1 | Damage status data, transmitted after damage occurs. | Main Control Module → respective robot | Data Link |
| 0x0207 | 7 | Real-time launch data, sent upon a projectile launched. | Main Control Module → respective robot | Data Link |
| 0x0208 | 6 | Projectile Allowance, transmitted at a fixed frequency of 10 Hz | Server → Own Side Hero, Infantry, Sentry, Drone | Data Link |
| 0x0209 | 5 | Robot RFID module status, sent at a fixed frequency of 3 Hz | Server → Own Side robots equipped with RFID modules | Data Link |
| 0x020A | 6 | Dart Player's Client Command data, sent at a fixed frequency of 3 Hz | Server → Own Side Dart Robot | Data Link |
| 0x020B | 40 | Ground robot location data, sent at a fixed frequency of 1 Hz | Server → Own Side Sentry | Data Link |
| 0x020C | 2 | Radar tracking progress data, transmitted at a fixed frequency of 1 Hz. | Server → Own Side Radar Robot | Data Link |
| 0x020D | 6 | Sentry decision data syncing, transmitted at a fixed rate of 1 Hz. | Server → Own Side Sentry | Data Link |
| 0x020E | 1 | Radar decision data synchronization, transmitted at a fixed frequency of 1 Hz. | Server → Own Side Radar Robot | Data Link |

| Command code | Data segment length | Description | Sender/Receiver | Data Link |
|---|---|---|---|---|
| 0x0301 | 127 | Interaction data between robots, transmission triggered by sender, with a maximum frequency of 30 Hz. | - | Data Link |
| 0x0302 | 30 | Custom Controller and robot Interaction data; transmission is triggered by the sender, with a maximum frequency of 30 Hz. | Custom Controller → robots connected to the player's client for video transmission | VTM Link |
| 0x0303 | 15 | Player's Client Mini-map interaction data, sent when triggered by the Player's Client | Player's Client click → Server → Own side robots selected by the Sender | Data Link |
| 0x0304 | 12 | Keyboard and mouse remote control data, sent at a fixed frequency of 30 Hz | Player's Client → robots connected via VTM to the Player's Client | VTM Link |
| 0x0305 | 24 | Player's Client Mini-map receives Radar data, with a maximum frequency of 5 Hz. | Radar → Server → all own side Player's Clients | Data Link |
| 0x0306 | 8 | Custom Controller and Player's Client interaction data, sent when triggered by the sender, with a maximum frequency of 30 Hz. | Custom Controller → Player's Client | - |
| 0x0307 | 103 | Player's Client Mini-map receives path data, with a maximum frequency of 1 Hz. | Sentry/semi-manually operated robots → respective Player's Client | Data Link |

| Command code | Data segment length | Description | Sender/Receiver | Data Link |
|---|---|---|---|---|
| 0x0308 | 34 | Player's Client mini-map receives data from robots at a maximum frequency of 3 Hz. | Own Side Robots → Own Side Player's Client | Data Link |
| 0x0309 | 30 | The Custom Controller receives data from robots at a maximum frequency of 10 Hz | Own side robots → Custom Controller connected to the corresponding Player's Client | VTM Link |
| 0x0310 | 150 | Data sent from robots to the Custom Client has a maximum frequency of 50 Hz. | Own side robots → VTM link → Custom Client connected to the corresponding Player's Client | VTM Link |
| 0x0F01 | • Send 1<br>• Receive 1 | Set the VTM Link output channel with a maximum frequency of 1 Hz. | • Send: Robots → VTM (Transmitter)<br>• Receive: VTM (Transmitter) → Robots | VTM Link |
| 0x0F02 | • Send 0<br>• Receive 1 | Query the current image output channel; maximum frequency is 2 Hz. | • Send: Robots → VTM (Transmitter)<br>• Receive: VTM (Transmitter) → Robots | VTM Link |

Table 1-5 0x0001

| Byte offset | Size | Description |
|---|---|---|
| 0 | 1 | bit 0–3: Competition Type |

| Byte offset | Size | Description |
|---|---|---|
| | | • 1: RoboMaster University Championship (RMUC)<br>• 2: Reserved<br>• 3: Reserved<br>• 4: RoboMaster University League (RMUL) 3v3 Match<br>• 5: RoboMaster University League (RMUL) Infantry Match<br>bit 4-7: Current match phase<br>• 0: Not started<br>• 1: Preparation Period<br>• 2: 15-Second Referee System Initialization Period<br>• 3: 5-Second Countdown<br>• 4: In Match<br>• 5: Match Settling |
| 1 | 2 | Remaining time in the current phase, in seconds |
| 3 | 8 | UNIX time becomes active once robots are properly connected to the Referee System's NTP Server. |

```
typedef _packed struct
{
  uint8_t game_type : 4;
  uint8_t game_progress : 4;
  uint16_t stage_remain_time;
  uint64_t SyncTimeStamp;
}game_status_t;
```

Table 1-6 0x0002

| Byte offset | Size | Description |
|---|---|---|
| 0 | 1 | • 0: Draw<br>• 1: Red Team wins<br>• 2: Blue Team wins |

```
typedef _packed struct
{
```

```
    uint8_t winner;
} game_result_t;
```

Table 1-7 0x0003

| Byte offset | Size | Description |
|---|---|---|
| 0 | 2 | Own Side Hero No. 1 HP; if this robot is not fielded or is ejected, its HP is 0. The same applies below. |
| 2 | 2 | Own Side Engineer No. 2 HP |
| 4 | 2 | Own Side Infantry No. 3 HP |
| 6 | 2 | Own Side Infantry No.4 HP |
| 8 | 2 | Reserved |
| 10 | 2 | Own Side Sentry No. 7 HP |
| 12 | 2 | Own Side Outpost HP |
| 14 | 2 | Own Side Base HP |

```
typedef _packed struct
{
   uint16_t Own Side 1 Infantry HP;
   uint16_t Own Side 2 Infantry HP;
   uint16_t Own Side 3 Infantry HP;
   uint16_t Own Side 4 Infantry HP;
   uint16_t reserved;
   uint16_t Own Side 7 Sentry HP;
   uint16_t Own Side Outpost HP;
   uint16_t Own Side Base HP;
} game_robot_HP_t;
```

Table 1-8 0x0101

| Byte offset | Size | Description |
|---|---|---|
| 0 | 4 | 0: Not occupied / not activated<br>1: Occupied / activated<br>● bit 0-2:<br>➢ bit 0: Occupation status of the own side Resupply Zone that does not overlap with the Resource Zone; 1 means occupied |

| Byte offset | Size | Description |
|---|---|---|
| | | ➢ bit 1: Occupation status of Own Side Resupply Zone overlapping with the Resource Zone; 1 means occupied<br>➢ bit 2: Occupation status of own side Resupply Zone; 1 indicates occupied (applicable to RMUL only)<br>● bit 3-6: Own Side Power Rune status<br>➢ bit 3-4: Activation status of Own Side Small Power Rune — 0 means not activated, 1 means activated, and 2 means activating.<br>➢ bit 5-6: Activation status of Own Side Large Power Rune; 0 means not activated, 1 means activated, and 2 means activating.<br>● bit 7-8: Occupation status of Own Side's central elevated ground, 1 means Occupied by Own Side, 2 means Occupied by Opponent.<br>● bit 9-10: Occupation status of Own Side Trapezoid-Shaped Elevated Ground; 1 indicates occupied<br>● bit 11-19: The time when the Opponent's Dart last hit Own Side's Outpost or Base (0-420, with an initial value of 0 at the start of the Round)<br>● bit 20-22: The specific goal where the Opponent's Dart last hit Own Side's Outpost or Base. The default value at the start of the Round is 0. 1 means the Outpost was hit, 2 means the fixed target in the Base was hit, 3 means a random fixed target in the Base was hit, 4 means a random moving target in the Base was hit, and 5 means the moving terminal target in the Base was hit.<br>● bit 23-24: Occupation status of the Central Buff Point—0 means not occupied, 1 means occupied by Own Side, 2 means occupied by Opponent, and 3 means occupied by both sides. (Only applicable to RMUL)<br>● bit 25-26: Occupation status of Own Side Fortress Buff Point: 0 means not occupied, 1 means occupied by Own Side, 2 means occupied by Opponent, and 3 means occupied by both sides.<br>● bit 27-28: Occupation status of Own Side Outpost Buff Point — 0 means not occupied, 1 means occupied by Own Side, 2 means occupied by Opponent.<br>● bit 29: Occupation status of Own Side Base Buff Point; 1 indicates occupied. |

| Byte offset | Size | Description |
|---|---|---|
| | | ● bit 30-31: Reserved |

```
typedef _packed struct
{
   uint32_t event_data;
} event_data_t;
```

Table 1-9 0x0104

| Byte offset | Size | Description |
|---|---|---|
| 0 | 1 | Own Side's most recent penalty level:<br>● 1: Dual Yellow Card<br>● 2: Yellow Card<br>● 3: Red Card<br>● 4: Forfeiture |
| 1 | 1 | ● The offending robot ID for Own Side's most recent penalty. (For example, the ID for Red 1 is 1, and for Blue 1 is 101.)<br>● For Forfeiture and when both sides receive a Yellow Card, this value is 0. |
| 2 | 1 | The number of violations at the corresponding Penalty Level committed by the Offending Robot in Own Side's most recent penalty. (Defaults to 0 at the beginning of the round.) ） |

```
typedef _packed struct
{
   uint8_t level;
   uint8_t offending_robot_id;
   uint8_t count;
}referee_warning_t;
```

Table 1-10 0x0105

| Byte offset | Size | Description |
|---|---|---|
| 0 | 1 | Own Side Dart launch remaining time, unit: seconds |
| 1 | 2 | ● bit 0-2: |

| Byte offset | Size | Description |
|---|---|---|
| | | The most recent Own Side Dart strike destination. At the start of the round, the default value is 0; 1 indicates a strike on the Outpost; 2 indicates a strike on the Base fixed target; 3 indicates a strike on the Base random fixed target; 4 indicates a strike on the Base random moving target; and 5 indicates a strike on the Base ternimal moving target.<br>● bit 3–5:<br>The opponent's most recently struck target, total strike count, default is 0 at the beginning of the round, with a maximum of 4.<br>● bit 6-7:<br>The strike destination currently selected for the dart: 0 if at the beginning of the round or when unselected/selected Outpost, 1 for selected Base fixed target, 2 for selected Base random fixed target, 3 for selected Base random moving target, and 4 for selected Base terminal moving target.<br>● bit 8-15: Reserved |

```
typedef _packed struct
{
   uint8_t dart_remaining_time;
   uint16_t dart_info;
}dart_info_t;
```

Table 1-11 0x0201

| Byte offset | Size | Description |
|---|---|---|
| 0 | 1 | Robot ID |
| 1 | 1 | Robot level |
| 2 | 2 | Current roboy HP |
| 4 | 2 | Maximum robot HP |
| 6 | 2 | Robot barrel heat cooling value per second |
| 8 | 2 | Maximum barrel heat |
| 10 | 2 | Maximum chassis power |

| Byte offset | Size | Description |
|---|---|---|
| 12 | 1 | Power Management Module output status:<br>● bit 0: Gimbal port output, 0 indicates no output, 1 indicates 24 V output<br>● bit 1: Chassis port output, 0 means no output, 1 means 24V output<br>● bit 2: Shooter port output, 0 indicates no output, 1 indicates 24V output |

```
typedef _packed struct
{    uint8_t robot_id;
   uint8_t robot_level;
   uint16_t current_HP;
   uint16_t maximum_HP;
   uint16_t shooter_barrel_cooling_value;
   uint16_t shooter_barrel_heat_limit;
   uint16_t chassis_power_limit;
   uint8_t power_management_gimbal_output : 1;
   uint8_t power_management_chassis_output : 1;
   uint8_t power_management_shooter_output : 1;
}robot_status_t;
```

1Table 1-12 0x0202

| Byte offset | Size | Description |
|---|---|---|
| 0 | 2 | Reserved |
| 2 | 2 | Reserved |
| 4 | 4 | Reserved |
| 8 | 2 | Buffer Energy (unit: J) |
| 10 | 2 | Barrel heat of the first 17mm Launching Mechanism |
| 12 | 2 | Barrel heat of the 42mm Launching Mechanism |

```
typedef _packed struct
{
   uint16_t reserved;
   uint16_t reserved;
   float reserved;
   uint16_t Buffer Energy;
   uint16_t Shooting Heat of the First 17 mm Launching Mechanism;
   uint16_t Shooting Heat of the 42 mm Launching Mechanism;
} power_heat_data_t;
```

Table 1-13 0x0203

| Byte offset | Size | Description |
|---|---|---|
| 0 | 4 | Current robot's x-coordinate, unit: m |
| 4 | 4 | Current robot's y-coordinate, unit: m |
| 8 | 4 | Orientation of the robot's Speed Monitor Module, unit: degree. Due north is 0 degree. |

```
typedef _packed struct
{
   float x;
   float y;
   float angle;
}robot_pos_t;
```

Table 1-14 0x0204

| Byte offset | Size | Description |
|---|---|---|
| 0 | 1 | HP Recovery Buff (percentage; a value of 10 indicates that 10% of Maximum HP is restored per second) |
| 1 | 2 | Specific value of barrel heat cooling buff (direct value; a value of x means cooling rate increases by x per second) |
| 3 | 1 | Defense Buff (percentage; a value of 50 indicates a 50% Defense Buff) |
| 4 | 1 | Negative Defense Buff (percentage; a value of 30 indicates a -30% Defense Buff) |
| 5 | 2 | Attack Buff (percentage; a value of 50 means a 50% Attack Buff) |
| 6 | 1 | bit 0–6: Feedback of remaining energy value for robots, represented as a hexadecimal proportion of the remaining energy. This is only reported when robots have less than 50% energy remaining; otherwise, the default feedback is 0x80. Robots are considered to start with 100% energy.<br>● bit 0: Set to 1 when remaining energy is ≥ 125%; otherwise, set to 0.<br>● bit 1: When remaining energy is ≥ 100%, the value is 1; otherwise, the value is 0.<br>● bit 2: Set to 1 when remaining energy is ≥ 50%, otherwise set to 0. |

| Byte offset | Size | Description |
|---|---|---|
| | | ● bit 3: Set to 1 when remaining energy is ≥ 30%, otherwise set to 0.<br>● bit 4: Set to 1 when remaining energy is at least 15%, otherwise set to 0.<br>● bit 5: Set to 1 when remaining energy is ≥ 5%; otherwise, set to 0.<br>● bit 6: Set to 1 when remaining energy is ≥ 1%; otherwise, set to 0. |

```
typedef _packed struct
{
   uint8_t recovery_buff;
   uint16_t cooling_buff;
   uint8_t defense_buff;
   uint8_t vulnerability_buff;
   uint16_t attack_buff;
   uint8_t remaining_energy;
} buff_t;
```

Table 1-15 0x0206

| Byte offset | Size | Description |
|---|---|---|
| 0 | 1 | bit 0-3: When the reason for HP Loss is that the projectile attack, collision, or disconnection, the value of these 4 bits represents the ID number of the Armor Module or Speed Monitor Module; if HP Loss is caused by miscellaneous reasons, the value is 0.<br>bit 4-7: HP variation type<br>● 0: HP loss caused by Armor Module being attacked by a projectile<br>● 1: HP loss caused by a disconnected Armor Module or Supercapacitor Management Module<br>● 5: HP loss occurs when the Armor Module suffers collision. |

```
typedef _packed struct
{
   uint8_t armor_id : 4;
   uint8_t HP_deduction_reason : 4;
} hurt_data_t;
```

The injury status for 0x0206 is determined locally by the Referee System and sent immediately. However, whether the corresponding damage is actually incurred depends on the rules and regulations. Please refer to the Server's final determination.

Table 1-16 0x0207

| Byte offset | Size | Description |
|---|---|---|
| 0 | 1 | Projectile type:<br>● bit 1: 17 mm Projectile<br>● bit 2: 42 mm Projectile |
| 1 | 1 | Launching Mechanism ID:<br>● 1: Launching Mechanism (17 mm)<br>● 2: Reserved<br>● 3: Launching Mechanism (42 mm) |
| 2 | 1 | Projectile firing rate (unit: Hz) |
| 3 | 4 | Projectile speed (unit: m/s) |

```
typedef _packed struct
{
   uint8_t projectile_type;
   uint8_t shooter_number;
   uint8_t launching_frequency;
   float projectile_speed;
}shoot_data_t;
```

Table 1-17 0x0208

| Byte offset | Size | Description |
|---|---|---|
| 0 | 2 | Projectile allowance of 17 mm projectiles |
| 2 | 2 | 42 mm projectile allowance |
| 4 | 2 | Remaining Gold Coin amount |
| 6 | 2 | Reserved 17mm Projectile Allowance provided by Fortress Buff Point;<br>This value is not related to whether the robot actually occupies the Fortress. |

```
typedef _packed struct
{
   uint16_t projectile_allowance_17mm;
   uint16_t projectile_allowance_42mm;
   uint16_t remaining_gold_coin;
   uint16_t projectile_allowance_fortress;
} projectile_allowance_t;
```

Table 1-18 0x0209

| Byte offset | Size | Description |
|---|---|---|
| 0 | 4 | The meaning of bit value 1/0: whether the Buff Point RFID card has been detected<br>● bit 0: Own Side Base Buff Point<br>● bit 1: Own Side Central Elevated Ground Buff Point<br>● bit 2: Opponent Central Elevated Ground Buff Point<br>● bit 3: Own Side Trapezoid-Shaped Elevated Ground Buff Point<br>● bit 4: Opponent Trapezoid-Shaped Elevated Ground Buff Point<br>● bit 5: Own Side Terrain Cross Buff Point (Launch Ramp) (near Own Side ramp, before the ramp)<br>● bit 6: Own Side Terrain Crossing Buff Point (Launch Ramp) (after the launch ramp on Own Side)<br>● bit 7: Opponent Terrain Crossing Buff Point (Launch Ramp) (Before Opponent-Side Launch Ramp)<br>● bit 8: Opponent Terrain Traversal Buff Point (Launch Ramp) (after the launch ramp on Opponent side)<br>● bit 9: Own Side Terrain Traversal Buff Point (below Trapezoid-Shaped Elevated Ground center)<br>● bit 10: Own Side Terrain Crossing Buff Point (above the central elevated ground)<br>● bit 11: Opponent Terrain Crossover Buff Point (below the central elevated ground)<br>● bit 12: Opponent Terrain Crossing Buff Point (Above Central Elevated Ground)<br>● bit 13: Own Side Terrain Crossing Buff Point (below the Road)<br>● bit 14: Own Side Terrain Crossing Buff Point (above the Road)<br>● bit 15: Opponent Terrain Crossing Buff Point (below Road)<br>● bit 16: Opponent Terrain Crossing Buff Point (above Road) |

| Byte offset | Size | Description |
|---|---|---|
| | | ● bit 17: Own Side Fortress Buff Point<br>● bit 18: Own Side Outpost Buff Point<br>● bit 19: Own Side Resupply Zone not overlapping with Resource Zone/RMUL Resupply Zone<br>● bit 20: Own Side Resupply Zone overlapping with Resource Zone<br>● bit 21: Own Side Assembly Buff Point<br>● bit 22: Opponent Assembly Buff Point<br>● bit 23: Central Buff Point (only applies to RMUL)<br>● bit 24: Opponent Fortress Buff Point<br>● bit 25: Opponent Outpost Buff Point<br>● bit 26: Own Side Terrain Crossing Buff Point (Tunnel) (Located below the Road Zone on Own Side)<br>● bit 27: Own Side Terrain Crossing Buff Point (Tunnel) (Located near the upper part of Own Side Road Zone)<br>● bit 28: Own Side Terrain Crossing Buff Point (tunnel) (near the lower area of Own Side Trapezoid-Shaped Elevated Ground)<br>● bit 29: Own Side Terrain Crossing Buff Point (tunnel) (near Own Side Trapezoid-Shaped Elevated Ground, higher area)<br>● bit 30: Opponent Terrain Crossing Buff Point (Tunnel) (Located near the lower area of Opponent's Road Zone)<br>● bit 31: Opponent Terrain Crossing Buff Point (Tunnel) (Located near the upper side of the Opponent Road Zone)<br>Note: All RFID cards are only active during the competition. Even outside of the match, the corresponding value remains 0 even if the associated RFID card is detected. |
| 4 | 1 | ● bit 0: Opponent Terrain Crossing Buff Point (tunnel) (near the lower end of Opponent Trapezoid-Shaped Elevated Ground) |

| Byte offset | Size | Description |
|---|---|---|
| | | ● bit 1: Opponent Terrain Crossing Buff Point (tunnel) (near the higher area of Opponent Trapezoid-Shaped Elevated Ground) |

```
typedef _packed struct
{
   uint32_t rfid_status;
   uint8_t rfid_status_2;
} rfid_status_t;
```

Table 1-19 0x020A

| Byte offset | Size | Description |
|---|---|---|
| 0 | 1 | Current status of the Dart Launching Station:<br>● 1: Off<br>● 2: Opening or closing<br>● 0: Already enabled |
| 1 | 1 | Reserved |
| 2 | 2 | Remaining competition time when switching strike target, measured in seconds; defaults to 0 if no switching action is performed. |
| 4 | 2 | Remaining time in the match when the operator last confirms the launch command, in seconds. The initial value is 0. |

```
typedef _packed struct
{
   uint8_t dart_launch_opening_status;
   uint8_t reserved;
   uint16_t target_change_time;
   uint16_t latest_launch_cmd_time;
}dart_client_cmd_t;
```

Table 1-20 0x020B

| Byte offset | Size | Description |
|---|---|---|
| 0 | 4 | Own Side Hero Location X-axis coordinate, unit: m |
| 4 | 4 | Own Side Hero Location Y-axis coordinate, unit: m |
| 8 | 4 | Own Side Engineer Location X-axis coordinate, unit: m |

| Byte offset | Size | Description |
|---|---|---|
| 12 | 4 | Own Side Engineer Location Y-axis coordinate, unit: m |
| 16 | 4 | Own Side Infantry No. 3 Location X-axis coordinate, unit: m |
| 20 | 4 | Own Side Infantry No. 3 Location Y-axis coordinate, unit: m |
| 24 | 4 | Own Side Infantry No. 4 Location X-axis coordinate, unit: m |
| 28 | 4 | Own Side Infantry No. 4 Location Y-axis, unit: m |
| 32 | 4 | Reserved |
| 36 | 4 | Reserved |

The intersection of the perimeter wall near the Red Team projectile supplier serves as the origin of the coordinate system. The X-axis extends along the long side of the match field toward the Blue Team, while the Y-axis extends along the short side toward the Red Team landing pad.

```
typedef _packed struct
{
   float hero_x;
   float hero_y;
   float engineer_x;
   float engineer_y;
   float infantry_3_x;
   float infantry_3_y;
   float infantry_4_x;
   float infantry_4_y;
   float reserved;
   float reserved;
}ground_robot_position_t;
```

Table 1-21 0x020C

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 2 | ● bit 0: Vulnerability status of Opponent Hero 1<br>● bit 1: Vulnerability status of Opponent Engineer No. 2<br>● bit 2: Vulnerability status of Opponent Infantry No. 3<br>● bit 3: Vulnerability status of Opponent Infantry No. 4<br>● bit 4: Vulnerability status of Opponent Sentry<br>● bit 5: Tracking status for Own Side Hero No. 1<br>● bit 6: Tracking status for Own Side Engineer No. 2<br>● bit 7: Tracking status of Own Side Infantry No. 3<br>● bit 8: Tracking status for Own Side Infantry Robot No. 4<br>● bit 9: Tracking status for Own Side Sentry<br>● bit 10-15: Reserved | ● Opponent Robots: Transmit "1" when the corresponding robot's Tracking Progress is greater than or equal to 100, and "0" when the Tracking Progress is less than 100.<br>● Own Side Robots: When the corresponding robot's Tracking Progress is ≥ 50, send 1; when Tracking Progress is < 50, send 0. |

```
typedef _packed struct
{
  uint16_t tracking_progress;
}radar_mark_data_t;
```

Table 1-22 0x020D

| Byte offset | Size | Description |
|---|---|---|
| 0 | 4 | ● bit 0-10: Except for Remote Exchange, the Projectile Allowance that the Sentry successfully exchanges; it starts at 0 at the beginning of the round, and after the Sentry successfully exchanges a certain Projectile Allowance, this value will change to match the Projectile Allowance exchanged by the Sentry.<br>● bit 11-14: The number of times the Sentry has successfully performed Remote Exchange for Projectile Allowance. This value starts at 0 at the beginning of the Round, and after the Sentry successfully completes Remote Exchange for Projectile Allowance, it will update to reflect the number of such exchanges performed by the Sentry.<br>● bit 15-18: Number of times the Sentry successfully performs Remote Exchange for HP. At the start of the Round, this value is 0; after the Sentry completes Remote Exchange for HP, the value updates to reflect the total successful Remote Exchange for HP actions.<br>● bit 19: Indicates whether the Sentry can confirm free respawn. The value is 1 when free respawn can be confirmed, otherwise it is 0.<br>● bit 20: Indicates whether Sentry can perform an Instant Respawn Exchange at this time. The value is 1 if Instant Respawn Exchange is available, otherwise it is 0.<br>● bit 21-30: The number of Gold Coins required for Sentry to Exchange for Instant Respawn at the current moment.<br>● bit 31: Reserved |
| 4 | 2 | ● bit 12-13: Current mode of Sentry; 1 indicates offensive, 2 indicates defensive, and 3 indicates mobile.<br>● bit 14: Indicates whether the Own Side Power Rune can enter the activating state; 1 means it is currently available.<br>● bit 15: Reserved |

```
typedef _packed struct
{
  uint32_t sentry_info;
```

```
  uint16_t sentry_info_2;
} sentry_info_t;
```

Table 1-23 0x020E

| Byte offset | Size | Description |
|---|---|---|
| 0 | 1 | ● bit 0-1: Indicates whether Radar has triggered double vulnerability; starts at 0. The value represents the number of opportunity available for Radar to trigger double vulnerability, up to a maximum of 2.<br>● bit 2: Whether the Opponent is currently triggered for double vulnerability<br>➢ 0: Opponent has not been triggered for double vulnerability<br>➢ 1: Opponent is currently being triggered for double vulnerability<br>● bit 3-4: Own Side Level (corresponding to Opponent Interference Difficulty Level), starts at 1, with a maximum of 3<br>● bit 5: Indicates whether the key can currently be modified; 1 means modification is allowed.<br>● bit 6–7: Reserved |

```
typedef _packed struct
{
  uint8_t radar_info;
} radar_info_t;
```

Robots interaction data is transmitted via the standard data link, and its data segment includes a unified data segment header structure. The data segment header structure includes the content ID, sender and receiver IDs, and the content data segment. The total length of a robot interaction data packet must not exceed 127 bytes. After subtracting 9 bytes for the frame_header, cmd_id, and frame_tail, as well as 6 bytes for the data segment header structure, the maximum size of the robot interaction content data segment is 112 bytes.

Within every 1000 milliseconds, Hero, Engineer, Infantry, Drone, and Dart can receive up to 3,720 bytes of data, while Radar and Sentry can receive up to 5,120 bytes.

There are multiple content IDs; however, the aggregate uplink update rate for cmd_id is limited to 30 Hz. Please schedule bandwidth usage accordingly.

Table 1-24 0x0301

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 2 | Sub-content ID | Must be an open sub-content ID |

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 2 | 2 | Sender ID | Must match Own Side ID; see appendix for ID numbers. |
| 4 | 2 | Receiver ID | ● Own Side communication only<br>● Must be an Inter-Robot Communication recipient permitted by the rules<br>● If the recipient is the Player's Client, messages can only be sent to the sender's corresponding Player's Client.<br>● ID numbers are detailed in the appendix. |
| 6 | x | Content data segment | Maximum x is 112. |

```
typedef _packed struct
{
  uint16_t data_cmd_id;
  uint16_t sender_id;
  uint16_t receiver_id;
  uint8_t user_data[x];
}robot_interaction_data_t;
```

| Sub-content ID | Length of the data segment | Instruction |
|---|---|---|
| 0x0200–0x02FF | x ≤ 112 | Communication between robots |
| 0x0100 | 2 | Player's Client deletes a layer |
| 0x0101 | 15 | Player's Client draws a figure |
| 0x0102 | 30 | Player's Client draws two figures |
| 0x0103 | 75 | Player's Client draws five figures |
| 0x0104 | 105 | Player's Client draws seven figures |
| 0x0110 | 45 | Player's Client draws character Figures |
| 0x0120 | 4 | Sentry Self-decision Command |
| 0x0121 | 1 | Radar Self-decision Command |

Table 1-25 Sub-content ID: 0x0100

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 1 | Delete operation | ● 0: No operation<br>● 1: Delete layer<br>● 2: Delete all |
| 1 | 1 | Number of layers | Number of layers: 0 to 9 |

```
typedef _packed struct
{
  uint8_t delete_type;
  uint8_t layer;
}interaction_layer_delete_t;
```

Table 1-26 Sub-content ID: 0x0101

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 3 | Figure name | Used as an index during operations such as Figure deletion and modification. |
| 3 | 4 | Figure Configuration 1 | bit 0-2: Figure operation<br>● 0: No operation<br>● 1: Add<br>● 2: Edit<br>● 3: Delete<br>bit 3-5: Figure type<br>● 0: Straight line<br>● 1: Rectangle<br>● 2: Perfect circle<br>● 3: Ellipse<br>● 4: Arc<br>● 5: Floating-point number |

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| | | | ● 6: Integer<br>● 7: Character<br>bit 6–9: Number of layers (0–9)<br>bit 10-13: Color<br>● 0: Red/Blue (Own Side color)<br>● 1: Yellow<br>● 2: Green<br>● 3: Orange<br>● 4: Magenta<br>● 5: Pink<br>● 6: Cyan<br>● 7: Black<br>● 8: White<br>bit 14-31: The meaning varies depending on the type of figure being drawn. For details, see "Table 1-27 Instruction for Figure Detail Parameters". |
| 7 | 4 | Figure Configuration 2 | bit 0-9: Line width. It is recommended that the font size to line width ratio be 10:1.<br>bit 10–20: Starting point/center x-coordinate<br>bit 21-31: Starting point/center y-coordinate |
| 11 | 4 | Figure Configuration 3 | Depending on the figure being drawn, the meaning varies. For details, see "Table 1-27 Instruction for Figure Detail Parameters" |

```
typedef _packed struct
{
  uint8_t figure_name[3];
  uint32_t operate_type:3;
  uint32_t figure_type:3;
  uint32_t layer:4;
```

```
    uint32_t color:4;
    uint32_t details_a:9;
    uint32_t details_b:9;
    uint32_t width:10;
    uint32_t start_x:11;
    uint32_t start_y:11;
    uint32_t details_c:10;
    uint32_t details_d:11;
    uint32_t details_e:11;
}interaction_figure_t;
```

Table 1-27 Instruction for Figure Detail Parameters

| Type | details_a | details_b | details_c | details_d | details_e |
|---|---|---|---|---|---|
| Straight line | - | - | - | End x coordinate | End y coordinate |
| rectangle | - | - | - | Diagonal vertex x coordinate | Diagonal vertex y coordinate |
| Perfect circle | - | - | Radius | - | - |
| Ellipse | - | - | - | x semi-axis length | y semi-axis length |
| Arc | Starting angle | End angle | - | x semi-axis length | y semi-axis length |
| Floating-point number | Font size | No effect | This value divided by 1,000 is the actual display value. | | |
| Integer | Font size | - | 32-bit integer, int32_t | | |
| Character | Font size | Character length | - | - | - |

- The angle value means that 0° points to the 12 o'clock direction and is drawn clockwise.
- Screen location: (0, 0) indicates the bottom-left corner, while (1920, 1080) indicates the top-right corner of the screen.

- Floating-point numbers: All integers are 32-bit. For floating-point numbers, the value displayed is the input value divided by 1,000. For example, if you enter 1234 in the bytes corresponding to details_c, details_d, or details_e, the Player's Client will actually display the value as 1.234.
- Even if the transmitted value exceeds the limits of the corresponding data type, the figure may still be displayed, but the display effect is not guaranteed.

Table 1-28 Subcontent ID: 0x0102

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 15 | Figure 1 | Same as the data segment of 0x0101. |
| 15 | 15 | Figure 2 | Same as the data segment of 0x0101. |

```
typedef _packed struct
{
  interaction_figure_t interaction_figure[2];
}interaction_figure_2_t;
```

Table 1-29 Subcontent ID: 0x0103

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 15 | Figure 1 | Same as the data segment of 0x0101. |
| 15 | 15 | Figure 2 | Same as the data segment of 0x0101. |
| 30 | 15 | Figure 3 | Same as the data segment of 0x0101. |
| 45 | 15 | Figure 4 | Same as the data segment of 0x0101. |
| 60 | 15 | Figure 5 | Same as the data segment of 0x0101. |

```
typedef _packed struct
{
  interaction_figure_t interaction_figure[5];
}interaction_figure_3_t;
```

Table 1-30 Sub-content ID: 0x0104

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 15 | Figure 1 | Same as the data segment of 0x0101. |
| 15 | 15 | Figure 2 | Same as the data segment of 0x0101. |
| 30 | 15 | Figure 3 | Same as the data segment of 0x0101. |
| 45 | 15 | Figure 4 | Same as the data segment of 0x0101. |
| 60 | 15 | Figure 5 | Same as the data segment of 0x0101. |
| 75 | 15 | Figure 6 | Same as the data segment of 0x0101. |
| 90 | 15 | Figure 7 | Same as the data segment of 0x0101. |

```
typedef _packed struct
{
  interaction_figure_t interaction_figure[7];
}interaction_figure_4_t;
```

Table 1-31 Sub-content ID: 0x0110

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 2 | Content ID of the data | 0x0110 |
| 2 | 2 | Sender's ID | Need to verify the sender's ID for accuracy |
| 4 | 2 | Receiver ID | The receiver's ID must be verified for accuracy; only sending to the Player's Client corresponding to the robot is supported. |
| 6 | 15 | Character configuration | For more details, see the figure data introduction. |
| 21 | 30 | Character | - |

```
typedef _packed struct
{
  graphic_data_struct_t grapic_data_struct;
  uint8_t data[30];
```

**} ext_client_custom_character_t;**

Table 1-32 Sentry Decision Command: 0x0120

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 4 | Sentry decision-related commands | ● bit 0: Whether the Sentry has confirmed respawn<br>➢ 0 means the Sentry confirms not to respawn, even if the Respawn Timer has finished.<br>➢ Release Notes<br>● bit 1: Sentry confirms Exchange for Instant Respawn or not<br>➢ 0 means the Sentry confirms not to exchange for Instant Respawn.<br>➢ 1 indicates that the Sentry confirmed to Exchange Instant Respawn. If the Sentry meets the requirements for Exchange Instant Respawn at this time, it will immediately spend Gold Coin for Exchange Instant Respawn.<br>● bit 2-12: The projectile amount value that the Sentry plans to Exchange, which starts at 0. After modifying this value, the Sentry can exchange for the Projectile Allowance at the Restoration Zone.<br>This value must increase monotonically; otherwise, it will be considered invalid.<br>**Example: This value can only be 0 at the beginning of the Round. Afterwards, the Sentry can change it from 0 to X, consuming X Gold Coin to successfully exchange for X Projectile Allowance. Afterward, Sentry can change it from X to X+Y, and so on.**<br>● bit 13-16: Number of times the Sentry requests Remote Exchange for Projectile Allowance. The initial value is 0, and updating this value allows the Sentry to request Remote Exchange for Projectile Allowance.<br>This value must increase monotonically, and each change can only increment by 1; otherwise, it will be considered invalid. |

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| | | | **Example: This value can only be 0 at the start of the Round. Afterwards, Sentry can change it from 0 to 1, which will consume Gold Coin for a Remote Exchange of Projectile Allowance. Afterward, Sentry can change the value from 1 to 2, and so on.**<br>• bit 17-20: Number of times Sentry requests remote Exchange of health points. The initial value is 0, and changing this value allows the Sentry to request remote exchange for HP.<br>This value must increase monotonically, and each change can only increment by 1; otherwise, it will be considered invalid.<br>**Example: This value can only be 0 at the start of the round. Afterwards, Sentry can change it from 0 to 1, which consumes Gold Coin for Remote Exchange of HP. Afterward, Sentry can change the value from 1 to 2, and so on.**<br>When Sentry sends this sub-command, the Server processes these Commands sequentially from lower to higher bits, until all are successful or cannot be processed further.<br>**Example: If the team's Gold Coin count is 0 and the Sentry is defeated, the value for "Confirm respawn" is 1, the value for "Conform Instant Respawn Exchange" is 1, and the value for "Confirm Exchange for Projectile Allowance" is 100. (Assuming the Sentry has not previously exchanged Projectile Allowance) Since the team's Gold Coin balance is insufficient for the Sentry to exchange for Instant Respawn, the Server will disregard subsequent commands and wait for the next set of commands sent by the Sentry.**<br>• bit 21-22: Sentry modifies its current mode command, with 1 for offensive, 2 for defensive, and 3 for mobile. The default is 3; changing this value updates the Sentry's mode.<br>• bit 23: Whether the Sentry confirms to set the Power Rune to activating status; 1 means yes. Default value is 0.<br>• bit 24-31: Reserved. |

```
typedef _packed struct
{
  uint32_t sentry_cmd;
  } sentry_cmd_t;
```

Table 1-33 Radar Decision Command: 0x0121

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 1 | Whether the Radar confirms to trigger double vulnerability | At the start of the round, the value is set to 0. You can request to trigger double vulnerability by modifying this value. If Radar has an opportunity to trigger double vulnerability at this time, it will be triggered.<br>This value must increase monotonically, and each change can only increment by 1; otherwise, it will be considered invalid.<br>**Example: This value must be 0 at the beginning of the Round. Afterwards, Radar can change it from 0 to 1. If Radar has a chance to trigger double vulnerability, double vulnerability will be triggered. Afterward, Radar can change the value from 1 to 2, and continue in this manner.**<br>If Radar requests double vulnerability while double vulnerability is already active, the second double vulnerability will become active once the first one ends. |
| 1 | 7 | Command for key update or verification | Each byte is encoded as an ASCII letter or number. Starts with a random value in each Round. byte1 indicates the Command type, while byte2-7 represent the key value.<br>When the value of byte1 is 1, modifying this value will update Own Side's encrypted key; when the value of byte1 is 2, modifying this value will transmit the Opponent's key, which Radar has decoded, to the Server to verify if it has been decoded correctly.<br>Note:<br>● You can only change the key at the beginning of the round, or when your Opponent successfully cracks it and the encryption Level (Own Side interference difficulty) increases. Any other attempts to modify the key will not be valid. |

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| | | | ● When byte1 is set to 2, any further updates within 10 seconds after the password is verified and updated will not be effective. |

```
typedef _packed struct
{
  uint8_t radar_cmd;
  uint8_t password_cmd;
  uint8_t password_1;
  uint8_t password_2;
  uint8_t password_3;
  uint8_t password_4;
  uint8_t password_5;
  uint8_t password_6;
} radar_cmd_t;
```

Table 1-34 Setting image transmission output channel: 0x0F01

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 1 | Receive 1 | Set the image transmission channel and receive feedback on the settings. | Set values 1–6: Assign the channel to 1 through 6.<br>Feedback value 0: Channel set successfully.<br>Feedback value 1: Figure transmission is starting, unable to set channel.<br>Feedback value 2: Channel setting error. Unable to set channel. |

Table 1-35 Setting Figure transmission output channel: 0x0F02

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0/1 | Send 0 | Query the Figure output channel. | No data segment is required; simply send this Command code to query the video transmission output channel.<br>Feedback value 0: Channel not set.<br>Feedback value 1–6: Indicates the currently active channel. |

# 1.3 Mini-map interaction data

## 1.3.1 Sending data via Player's Client

- The Gimbal Operator can use the Player's Client main map to send predefined data to the robot.

Command code is 0x0303, transmitted when triggered, with a minimum interval of 0.5 seconds between each transmission.

**Sending method 1:**

① Click on the Own Side robot's icon.

② (Optional) Press a key on the keyboard, or click the Opponent robot's avatar;

③ Click anywhere on the Mini-map. This method sends map coordinate data to the selected Own Side robot; if you click on an Opponent Robot avatar, the target robot's ID will be sent instead of the coordinate data.

**Sending Method 2:**

① (Optional) Press a keyboard key or click the Opponent robot's avatar.

② Click anywhere on the Mini-map. This method sends map coordinate data to all robots on Own Side. If you click on the Opponent robot's avatar, the target robot ID will replace the coordinate data

- Operators corresponding to robots in semi-automatic control mode can send preset data to Robots via the Player's Client main map.

The command code is 0x0303. It is sent when triggered, and the interval between two transmissions must not be less than 3 seconds.

**Sending Method:**

① (Optional) Press a keyboard key or click the Opponent robot's avatar.

② Click anywhere on the Mini-map. This method sends map coordinate data to the robot associated with the Operator. If the Opponent's robot avatar is clicked, the target Robot ID will replace the coordinate data.

A single robot in semi-automatic control mode can receive information sent by the Gimbal Operator as well as by the corresponding Operator. The sources of the two types of information are distinguished in the "Information Source" column in the table below.

To reduce the impact of occasional instability in robot serial port receiver devices on communication, the sending mechanism for the 0x0303 protocol has a special arrangement: after the Player's Client triggers a single transmission, the Server will send the same message to the robots an additional four times at 100 ms intervals, for a total of five transmissions. Afterwards, until the next time the Player's

Client triggers a transmission, the Server will continue to send the most recent packet at a fixed frequency of 1 Hz. The timing for continuous sending upon trigger and the timing for fixed 1 Hz frequency transmission operate independently. Teams need to pay attention to how repeatedly received duplicate protocol packets are handled.

Table 1-36 Command ID: 0x0303

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 4 | X-axis coordinate of the target location, in meters | When sending the Robot ID, this value is 0. |
| 4 | 4 | y-axis coordinate of the target location, in meters | When sending the Robot ID, this value is 0. |
| 8 | 1 | General key value for the keyboard key pressed by the Gimbal Operator | If no key is pressed, the value is 0. |
| 9 | 1 | Opponent Robot ID | When sending coordinate data, this value is 0. |
| 10 | 2 | Information source ID | ID of the information source, with detailed ID mapping provided in the appendix |

```
typedef _packed struct
{
  float opponent_position_x;
  float opponent_position_y;
  uint8_t cmd_keyboard;
  uint8_t opponent_robot_id;
  uint16_t source_id;
}map_command_t;
```

## 1.3.2 Receiving data via Player's Client

The Mini-map on the Player's Client can receive data from robots.

Radar can transmit the coordinates of opponent robots to all Player's Clients on Own Side via a standard data link, and these locations will be displayed on the Mini-map of Own Side Player's Clients.

Table 1-37 Command ID: 0x0305

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 2 | Hero x coordinate, unit: cm | If the x or y values exceed the boundaries, the point will appear at the corresponding edge. If both x and y are zero, it is considered that the robot's location data has not been sent. |
| 2 | 2 | Hero y location coordinate, unit: cm | |
| 4 | 2 | Engineer x location coordinate, unit: cm | |
| 6 | 2 | Engineer y location coordinate, unit: cm | |
| 8 | 2 | Infantry No. 3 x location coordinate, unit: cm | |
| 10 | 2 | Infantry No. 3 y Location coordinate, unit: cm | |
| 12 | 2 | Infantry No. 4 x location coordinate, unit: cm | |
| 14 | 2 | Infantry No. 4 y location coordinate, unit: cm | |
| 16 | 2 | Infantry No. 5 x location coordinate, unit: cm | |
| 18 | 2 | Infantry No. 5 y location coordinate, unit: cm | |
| 20 | 2 | Sentry x location coordinate, unit: cm | |
| 22 | 2 | Sentry y location coordinate, unit: cm | |

```
typedef _packed struct
{
  uint16_t hero_location_x;
  uint16_t hero_location_y;
  uint16_t engineer_location_x;
  uint16_t engineer_position_y;
  uint16_t infantry_3_position_x;
  uint16_t infantry_3_position_y;
```

```
  uint16_t infantry_4_position_x;
  uint16_t infantry_4_position_y;
  uint16_t infantry_5_position_x;
  uint16_t infantry_5_position_y;
  uint16_t sentry_position_x;
  uint16_t sentry_position_y;
} map_robot_data_t;
```

Sentry or semi-automated robots can send path coordinate data via a standard data link to the corresponding Player's Client, and the path will be displayed on the Mini-map.

Table 1-38 Command Code ID: 0x0307

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 1 | 1: Move to the designated location and attack<br>2: Move to the designated location and defend<br>3: Move to the designated location | - |
| 1 | 2 | Starting x-axis coordinate of the path, unit: dm | The bottom-left corner of the mini-map is the origin of the coordinate system, with the positive X-axis extending to the right horizontally and the positive Y-axis extending upward vertically. The display location will be scaled proportionally according to the match area size and mini-map size, and any location outside the boundaries will be shown at the edge. |
| 3 | 2 | Starting y-axis coordinate of the path, unit: dm | |
| 5 | 49 | Array of X-axis increments for path points, unit: dm | Incremental values are calculated relative to the previous position, with 49 new positions in total. Each position is determined by corresponding increments along the X and Y axes. |
| 54 | 49 | Array of y-axis increments for path points, unit: dm | |
| 103 | 2 | Sender ID | Must match Own Side ID; see appendix for ID codes. |

```
typedef _packed struct
```

```
{
   uint8_t intention;
   uint16_t start_position_x;
   uint16_t start_position_y;
   int8_t delta_x[49];
   int8_t delta_y[49];
   uint16_t sender_id;
}map_data_t;
```

Own Side robots can use a standard data link to send customized messages to any Own Side Player's Client, and these messages will be displayed at a specific location on the Own Side Player's Client.

Table 1-39 Command Code ID: 0x0308

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 2 | Sender ID | Need to verify the sender's ID for accuracy |
| 2 | 2 | Receiver ID | The receiver's ID must be validated for accuracy; data can only be sent to Own Side Player's Client. |
| 4 | 30 | Character | Encoded in UTF-16 format for transmission, allowing Chinese characters to be displayed. When encoding and sending data, please pay attention to the endianness issue. |

```
typedef _packed struct
{
   uint16_t sender_id;
   uint16_t receiver_id;
   uint8_t user_data[30];
 } custom_info_t;
```

# 1.4 VTM link data instruction

## 1.4.1 Instruction for custom controller and robots interaction data

Operators can use a custom controller to send data to the corresponding robots via the VTM Link.

Table 1-40 Command Code ID: 0x0302

| Byte offset | Size | Description |
|---|---|---|
| 0 | 30 | Custom Data |

```
typedef _packed struct
{
   uint8_t data[x];
}custom_robot_data_t;
```

Robots can transmit data via VTM link to the custom controller connected to the corresponding Player's Client (not applicable to RMUL).

Table 1-41 Command ID: 0x0309

| Byte offset | Size | Description |
|---|---|---|
| 0 | 30 | Custom Data |

```
typedef _packed struct
{
   uint8_t data[x];
}robot_custom_data_t;
```

Table 1-42 Command ID: 0x0310

| Byte offset | Size | Description |
|---|---|---|
| 0 | 150 | Custom Data |

```
typedef _packed struct
{
        uint8_t data[x];
}robot_custom_data_2_t;
```

## 1.4.2 Keyboard and mouse remote control data

Keyboard and mouse control data sent via the remote controller will be simultaneously transmitted to the corresponding robots through the VTM link.

Table 1-43 Command ID: 0x0304

| Byte offset | Size | Description |
|---|---|---|
| 0 | 2 | Mouse x-axis movement speed; a negative value indicates movement to the left |
| 2 | 2 | Mouse y-axis movement speed; negative values indicate movement downward. |
| 4 | 2 | Mouse scroll wheel speed; negative values indicate scrolling backward |
| 6 | 1 | Whether the left mouse button is pressed: 0 means not pressed, 1 means pressed. |
| 7 | 1 | Mouse right button pressed: 0 means not pressed, 1 means pressed |
| 8 | 2 | Keyboard key information: each bit corresponds to a key, with 0 indicating the key is not pressed, and 1 indicating the key is pressed:<br>● bit 0: W key<br>● bit 1: S key<br>● bit 2: A key<br>● bit 3: D key<br>● bit 4: Shift key<br>● bit 5: Ctrl key<br>● bit 6: Q key<br>● bit 7: E key<br>● bit 8: R key<br>● bit 9: F key<br>● bit 10: G key<br>● bit 11: Z key<br>● bit 12: X key<br>● bit 13: C key<br>● bit 14: V key<br>● bit 15: B key |

| Byte offset | Size | Description |
|---|---|---|
| 10 | 2 | Reserved |

```
typedef _packed struct
{
  int16_t mouse_x;
  int16_t mouse_y;
  int16_t mouse_z;
  int8 left_button_down;
  int8 right_button_down;
  uint16_t keyboard_value;
  uint16_t reserved;
}remote_control_t;
```

## 1.5 Instructions for non-data link data

The Operator can use a custom controller to simulate keyboard and mouse operations on the Player's Client.

Table 1-44 Command Code ID: 0x0306

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| 0 | 2 | Keyboard key values:<br>● bit 0-7: Key value of Button 1<br>● bit 8-15: Key 2 value | ● Only responds to keys enabled on the Player's Client.<br>● Use general key values, supports 2-key rollover, changing the key value order will not affect the pressed state, and if no new key information is received, the pressed state from the previous frame will be maintained. |
| 2 | 2 | ● bit 0-11: Mouse x-axis pixel location<br>● bit 12-15: Mouse left button status | ● Location information uses absolute pixel values (the competition client operates at a resolution of 1920×1080, with the top-left corner of the screen as (0, 0)). |
| 4 | 2 | ● bit 0-11: Mouse y-axis pixel location<br>● bit 12-15: Right mouse button status | |

| Byte offset | Size | Description | Remarks |
|---|---|---|---|
| | | | ● When the mouse button status is 1, it indicates the button is pressed; any other value means it is not pressed. This information is only updated when the mouse icon appears. If there is no new mouse data, the Player's Client will retain the mouse information from the previous frame. Once the mouse icon disappears, this data will no longer be retained. |
| 6 | 2 | Reserved | - |

To perform a mouse move-and-click action, first send a data frame indicating the mouse is not pressed at the specified location, then send a data frame indicating the mouse is pressed while holding that location, and finally send a data frame indicating the mouse is not pressed while still holding that location.

```
typedef _packed struct
{
  uint16_t key_value;
  uint16_t x_location:12;
  uint16_t mouse_left:4;
  uint16_t y_location:12;
  uint16_t mouse_right:4;
  uint16_t reserved;
}custom_client_data_t;
```

# 2. Custom client protocol

The data stream format for the custom client uses Protobuf v3 (Protocol Buffers), with MQTT as the message publishing and subscription protocol. MQTT transmits the Protobuf-serialized binary stream directly. The topic corresponds to the command name. The server IP address is set to a fixed IP: 192.168.12.1, and the port is 3333. Custom Client IP addresses are automatically assigned via DHCP.

The communication process is as follows:

1. Write proto files based on the Protobuf message format in the documentation, and use protoc to generate class code in various languages.
2. On the publisher side, serialize the object into binary.
3. Send the binary data to the corresponding topic via MQTT publish.
4. On the subscriber side, receive messages and deserialize them using Protobuf.

Sample code is provided in “Appendix III: Sample Communication Code for Custom Client”.

In addition, a custom client can access the video transmission stream data by listening to port 3334 via UDP. The encoding format is HEVC, and the first 8 bytes of the stream data in each UDP packet are fixed as follows:

Frame number (incremental): 2 byte

Fragment index within current frame: 2 byte

Total bytes in current frame: 4 byte

## 2.1 Command overview

Table 2-45 Command overview

| Command (message) name | Command purpose | Sender/Receiver | Maximum qos Level | Frequency |
|---|---|---|---|---|
| RemoteControl | Transmit mouse and keyboard input, as well as custom data | Custom Client→Server | 1 | 75 Hz |
| GameStatus | Synchronize the global round status information | Server → Custom Client | 1 | 5 Hz |
| GlobalUnitStatus | Sync the status of Base, Outpost, and all robots | Server → Custom Client | 1 | 1 Hz |

| | | | | |
|---|---|---|---|---|
| GlobalLogisticsStatus | Synchronize global logistics information | Server → Custom Client | 1 | 1 Hz |
| GlobalSpecialMechanism | Synchronize active global special mechanism | Server → Custom Client | 1 | 1 Hz |
| Event | Global round event notification | Server → Custom Client | 1 | Sending upon trigger |
| RobotInjuryStat | Cumulative damage during one alive period | Server → Custom Client | 1 | 1 Hz |
| RobotRespawnStatus | Robots respawn status synchronization | Server → Custom Client | 1 | 1 Hz |
| RobotStaticStatus | Robots' fixed attributes and configurations | Server → Custom Client | 1 | 1 Hz |
| RobotDynamicStatus | Real-time robot data | Server → Custom Client | 1 | 10 Hz |
| RobotModuleStatus | Operating status of each module of the robots | Server → Custom Client | 1 | 1 Hz |
| RobotPosition | Robot spatial coordinates and orientation | Server → Custom Client | 1 | 1 Hz |
| Buff | Buff effect information | Server → Custom Client | 1 | Triggered to send upon gaining a buff, then sent at a fixed frequency of 1 Hz until the buff ends. |
| PenaltyInfo | Penalty information synchronization | Server → Custom Client | 1 | Sending upon trigger |
| RobotPathPlanInfo | Sentinel trajectory planning information | Server → Custom Client | 1 | 1 Hz |

| | | | | |
|---|---|---|---|---|
| MapClickInfoNotify | Gimbal Operator Map Click Marker | Custom Client→Server | 1 | Sending upon trigger |
| RadarInfoToClient | Location information of robots sent by Radar | Server → Custom Client | 1 | 1 Hz |
| CustomByteBlock | Custom data stream uploaded by robots (corresponding to 0x0310 on the Onboard Modules) | Robots → VTM Link → Custom Client | 1 | 50 Hz |
| AssemblyCommand | Engineer Assembly Command | Custom Client→Server | 1 | 1 Hz |
| TechCoreMotionStateSync | Tech Core motion state sync | Server → Custom Client | 1 | 1 Hz |
| RobotPerformanceSelectionCommand | Infantry/Hero performance system selection | Custom Client→Server | 1 | 1 Hz |
| RobotPerformanceSelectionSync | Infantry/Hero performance system status sync | Server → Custom Client | 1 | 1 Hz |
| HeroDeployModeEventCommand | Commands related to Hero Deploy Mode | Custom Client→Server | 1 | 1 Hz |
| DeployModeStatusSync | Deploy mode status synchronization | Server → Custom Client | 1 | 1 Hz |
| RuneActivateCommand | Power Rune activation command | Custom Client→Server | 1 | 1 Hz |
| RuneStatusSync | Power Rune status sync | Server → Custom Client | 1 | 1 Hz |
| SentinelStatusSync | Sentry pose status sync | Server → Custom Client | 1 | 1 Hz |
| DartCommand | Dart Control Command | Custom Client→Server | 1 | 1 Hz |
| DartSelectControlStatusSync | Dart selection status synchronization | Server → Custom Client | 1 | 1 Hz |

| | | | | |
|---|---|---|---|---|
| GuardCtrlCommand | Sentry control command request | Custom Client→Server | 1 | 1 Hz |
| GuardCtrlResult | Feedback on Sentry Control commands | Server → Custom Client | 1 | 1 Hz |
| AirSupportCommand | Air support command | Custom Client→Server | 1 | 1 Hz |
| AirSupportStatusSync | Air support status feedback | Server → Custom Client | 1 | 1 Hz |

# 2.2 Detailed protocol specification

## 2.2.1 RemoteControl

Purpose: transmit mouse and keyboard inputs, and custom data

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | int32 | Mouse x-axis movement speed, with negative values indicating movement to the left |
| 2 | int32 | Mouse y-axis movement speed, with negative values indicating downward movement |
| 3 | int32 | Mouse scroll wheel speed; negative values indicate scrolling backward |
| 4 | bool | Whether the left mouse button is pressed (false = released, true = pressed) |
| 5 | bool | Is the right mouse button pressed (false = released, true = pressed) |
| 6 | uint32 | Keyboard key bitmask |
| 7 | bool | Is the middle button pressed (false = released, true = pressed) |
| 8 | bytes | Up to 30 bytes of custom data |

```
message RemoteControl {
int32 mouse_x = 1;
int32 mouse_y = 2;
int32 mouse_z = 3;
bool left_button_down = 4;
bool right_button_down = 5;
uint32 keyboard_value = 6;
bool mid_button_down = 7;
bytes data = 8;
}
```

## 2.2.2 GameStatus

Purpose: Synchronize the overall competition status information.

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Current round number (starting from 1) |
| 2 | uint32 | Total number of rounds |
| 3 | uint32 | Red Team score |
| 4 | uint32 | Blue Team score |
| 5 | uint32 | Current phase |
| 6 | int32 | Remaining time in current phase (seconds) |
| 7 | int32 | Elapsed time in current stage (seconds) |
| 8 | bool | Is the match paused |

Enumeration values for current_stage:

- 0: Match Not Started
- 1: Preparation Period
- 2: 15-Second Referee System Initialization Period
- 3: 5-Second Countdown
- 4: In Match
- 5: Match Settling

```
message GameStatus {
uint32 current_round = 1;
uint32 total_rounds = 2;
uint32 red_score = 3;
uint32 blue_score = 4;
uint32 current_stage = 5;
int32 stage_countdown_sec = 6;
int32 stage_elapsed_sec = 7;
bool is_paused = 8;
}
```

## 2.2.3 GlobalUnitStatus

Purpose: Synchronize the statuses of the Base, Outpost, and all robots.

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Current Base HP (0 = destroyed) |
| 2 | uint32 | Base status |
| 3 | uint32 | Current Base shield value (0 = no shield) |

| Data ID | Data type | Data purpose |
|---|---|---|
| 4 | uint32 | Current Outpost HP (0 = destroyed) |
| 5 | uint32 | Outpost status |
| 6 | repeated uint32 | All robots' HP (own side first, then opponent) |
| 7 | repeated int32 | Own Side robots' remaining total projectile count |
| 8 | uint32 | Own Side's total accumulated damage |
| 9 | uint32 | Opponent's total accumulated damage |

Enumeration of base_status values:

- 0: Invincible
- 1: Invincibility removed, armor not deployed
- 2: Invincibility removed, armor deployed

Enumeration of outpost_status values:

- 0: Invincible
- 1: Alive, invincibility removed, center armor rotating
- 2: Alive, invincibility removed, central armor halted
- 3: Destroyed, cannot be rebuilt
- 4: Destroyed, can be rebuilt

```
message GlobalUnitStatus {
uint32 base_health = 1;
uint32 base_status = 2;
uint32 base_shield = 3;
uint32 outpost_health = 4;
uint32 outpost_status = 5;
repeated uint32 robot_health = 6;
repeated int32 robot_ammo = 7;
uint32 total_damage_red = 8;
uint32 total_damage_blue = 9;
}
```

## 2.2.4 Global Logistics Status

Purpose: Synchronize global logistics information

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Own Side current economy |
| 2 | uint64 | Own Side total accumulated economy |
| 3 | uint32 | Own Side Tech Core Level (The highest assembly difficulty level previously accomplished by the Engineer) |

| Data ID | Data type | Data purpose |
|---|---|---|
| 4 | uint32 | Own Side encryption level (Radar signal information mechanism) |

```
message GlobalLogisticsStatus {
uint32 remaining_economy = 1;
uint64 total_economy_obtained = 2;
uint32 own_side_level = 3;
uint32 encryption_level = 4;
}
```

## 2.2.5 GlobalSpecialMechanism

Purpose: Synchronize globally active special mechanisms.

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | repeated uint32 | List of active mechanism IDs |
| 2 | repeated int32 | Corresponding time parameters (seconds) |

Enumeration of mechanism_id values:

- 1: Timer for Own Side Fortress being occupied by Opponent
- 2: Timer for Opponent Fortress being occupied by Own Side

```
message GlobalSpecialMechanism {
repeated uint32 mechanism_id = 1;
repeated int32 mechanism_time_sec = 2;
}
```

## 2.2.6 Event

Purpose: Global event notification

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | int32 | Event ID |
| 2 | string | Event parameters (meaning varies depending on event_id) |

Enumeration of event_id values:

- 1: Elimination event (parameter: concatenation of the eliminator's ID and the destroyed robots' ID)

- 2: Base or Outpost destruction event (parameter value is the destroyed target ID, e.g., Blue Team Outpost is 111)
- 3: Power Rune available statuses count change (parameter value: number of available statuses after change)
- 4: The Power Rune currently enters the activating status (no parameter value)
- 5: Number of light arms successfully activated on the current Power Rune, and the average ring count (parameter value is the number of activated arms plus the average ring count)
- 6: Power Rune activated (including activation type)
- 7: Own Side Hero enters deploy mode (no parameter value)
- 8: Own Side Hero inflicts sniper damage (parameter is the total sniper damage inflicted)
- 9: Opponent Hero dealt sniper damage (parameter value is the total sniper damage inflicted)
- 10: Own Side calls for air support (no parameters)
- 11: Own Side air support interrupted (parameter value indicates the remaining number of interruptions available to the Opponent)
- 12: Opponent calls for air support (no parameter value)
- 13: Opponent air support interrupted (parameter value: remaining number of interruptions available for Own Side)
- 14: Dart Hit (parameter indicates the target hit: 1 for Outpost, 2 for Base fixed target, 3 for Base random fixed target, 4 for Base random moving target, 5 for Base ternimal moving target)
- 15: Both sides' dart gates open (parameter value 1 indicates Own Side's gate opened, 2 indicates Opponent's gate opened)
- 16: Own Side Base is under attack (no parameter value; each trigger has a built-in 5 s cooldown)
- 17: Both Outposts have stopped rotating (parameter value 1 for Own Side Outpost, 2 for Opponent Outpost).
- 18: Base Protective Armor deployed for both sides (parameter value 1 indicates Own Side Base, 2 indicates Opponent Base)

```
message Event {
int32 event_id = 1;
string param = 2;
}
```

## 2.2.7 RobotInjuryStat

Purpose: Cumulative damage statistics for robots during a single alive period

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Total damage accumulated during this alive period |
| 2 | uint32 | Collision damage |
| 3 | uint32 | 17 mm Projectile damage |
| 4 | uint32 | 42 mm projectile damage |
| 5 | uint32 | Dart splash damage |
| 6 | uint32 | Module Offline HP Loss |
| 7 | uint32 | WiFi Offline HP Loss |
| 8 | uint32 | Penalty HP Loss |
| 9 | uint32 | Server forces HP Loss by marking as defeated. |

| Data ID | Data type | Data purpose |
|---|---|---|
| 10 | uint32 | Destroyer ID (if no destroyer is detected, the value is 0) |

```
message RobotInjuryStat {
uint32 total_damage = 1;
uint32 collision_damage = 2;
uint32 small projectile damage = 3;
uint32 large projectile damage = 4;
uint32 dart splash damage = 5;
uint32 module_offline_damage = 6;
uint32 wifi_offline_damage = 7;
uint32 penalty_damage = 8;
uint32 server_kill_damage = 9;
uint32 killer_id = 10;
}
```

## 2.2.8 Robot Revive Status

Purpose: Robots Revive status synchronization

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | bool | Is the robot in pending respawn state |
| 2 | uint32 | Total Respawn Timer required for respawn |
| 3 | uint32 | Current Respawn Timer progress |
| 4 | bool | Is free respawn available |
| 5 | uint32 | Gold Coin cost required to respawn |
| 6 | bool | Is spending Gold Coin for respawn allowed? |

```
message RobotRespawnStatus {
bool is_pending_revive = 1;
uint32 total_revive_progress = 2;
uint32 current_revive_progress = 3;
bool can_revive_for_free = 4;
uint32 gold_coin_cost_for_revive = 5;
bool can_pay_for_Revive = 6;
}
```

## 2.2.9 Robot Static Attributes and Configuration

Purpose: Robots' fixed attributes and configuration

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Connection status (0 = not connected, 1 = connected) |
| 2 | uint32 | Field status (0 = Entered the field, 1 = Not entered) |
| 3 | uint32 | Alive Status (0 = Unknown, 1 = Alive, 2 = Defeated) |
| 4 | uint32 | Robot numbering |
| 5 | uint32 | Robot type |
| 6 | uint32 | Performance System - Launching Mechanism |
| 7 | uint32 | Performance System - Chassis |
| 8 | uint32 | Current level |
| 9 | uint32 | Maximum HP |
| 10 | uint32 | Maximum Heat |
| 11 | float | Barrel Heat Cooling Rate (per second) |
| 12 | uint32 | Maximum Power |
| 13 | uint32 | Maximum Buffer Energy |
| 14 | uint32 | Maximum Chassis Energy |

Enumeration of performance system shooter values:

- 1: Cooling-focused
- 2: Burst-focused
- 3: Hero Melee-focused
- 4: Hero Sniper-focused

performance_system_chassis enumeration values:

- 1: HP-focused
- 2: Power-focused
- 3: Hero Melee-focused
- 4: Hero Sniper-focused

```
message RobotStaticStatus {
uint32 connection_state = 1;
uint32 field_state = 2;
uint32 alive_state = 3;
uint32 robot_id = 4;
uint32 robot_type = 5;
uint32 performance_system_shooter = 6;
uint32 performance_system_chassis = 7;
uint32 level = 8;
uint32 max_health = 9;
uint32 max heat = 10;
float heat cooldown rate = 11;
```

```
uint32 max_power = 12;
uint32 max_buffer_energy = 13;
uint32 max_chassis_energy = 14;
}
```

## 2.2.10 Robots Dynamic Status

Purpose: Real-time data for robots

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Current HP |
| 2 | float | Current heat |
| 3 | float | Previous projectile velocity |
| 4 | uint32 | Current remaining Chassis energy |
| 5 | uint32 | Current Buffer Energy |
| 6 | uint32 | Current Experience Point |
| 7 | uint32 | Experience required to level up |
| 8 | uint32 | Total projectiles fired |
| 9 | uint32 | Remaining Projectile Allowance |
| 10 | bool | Is Out-of-Combat status active |
| 11 | uint32 | Countdown to Out-of-Combat status |
| 12 | bool | Is remote HP restoration available |
| 13 | bool | Is remote reload available |

```
message RobotDynamicStatus {
uint32 current_health = 1;
float current_heat = 2;
float last projectile firing rate = 3;
uint32 current chassis energy = 4;
uint32 current buffer energy = 5;
uint32 current_experience = 6;
uint32 experience_needed_for_upgrade = 7;
uint32 total_projectiles_fired = 8;
uint32 remaining_ammo = 9;
bool is_out_of_combat = 10;
uint32 out_of_combat_countdown = 11;
bool can_be_repaired_remotely = 12;
bool can_remote_ammo = 13;
}
```

## 2.2.11 RobotModuleStatus

Purpose: Operating status of each module in Robots

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Power Management Module status (0 = Offline, 1 = Online) |
| 2 | uint32 | RFID module status (0 = Offline, 1 = Online) |
| 3 | uint32 | Light Indicator Module status (0 = Offline, 1 = Online) |
| 4 | uint32 | 17 mm Launching Mechanism status (0 = Offline, 1 = Online) |
| 5 | uint32 | 42 mm Launching Mechanism status (0 = Offline, 1 = Online) |
| 6 | uint32 | UWB Positioning Module status (0 = Offline, 1 = Online) |
| 7 | uint32 | Armor Module Status (0 = Offline, 1 = Online) |
| 8 | uint32 | Video Transmission Module Status (0 = Offline, 1 = Online) |
| 9 | uint32 | Capacitor Module Status (0=Offline, 1=Online) |
| 10 | uint32 | Main Controller status (0 = Offline, 1 = Online) |

```
message RobotModuleStatus {
uint32 power_manager = 1;
uint32 RFID = 2;
uint32 light_strip = 3;
uint32 small_shooter = 4;
uint32 big_shooter = 5;
uint32 uwb = 6;
uint32 armor = 7;
uint32 video transmission = 8;
uint32 capacitor = 9;
uint32 main_controller = 10;
}
```

## 2.2.12 RobotPosition

Purpose: Robots' spatial coordinates and orientation

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | float | World coordinate X-axis |
| 2 | float | World coordinate Y-axis |
| 3 | float | World coordinate Z-axis |
| 4 | float | The orientation of this robot's Speed Monitor Module, unit: degree, with true north set as 0 degree. |

```
message RobotPosition {
```

```
float x = 1;
float y = 2;
float z = 3;
float yaw = 4;
}
```

## 2.2.13 Buff

Purpose: Buff effect information

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Robot ID |
| 2 | uint32 | Buff type |
| 3 | int32 | Buff value |
| 4 | uint32 | Maximum remaining time for buff |
| 5 | uint32 | Buff remaining time |
| 6 | string | Additional text parameter |

Enumeration of buff type values:

- 1: Attack Buff
- 2: Defense Buff
- 3: Barrel Heat Cooling Buff
- 4: Chassis Power Buff
- 5: HP Recovery Buff
- 6: Exchangeable Projectile Allowance
- 7: Terrain Crossing Buff (Preparation) (refers to scenarios where the first half of the Launch Ramp's RFID Interaction Module is detected, but the actual terrain crossing buff from the Launch Ramp has not been obtained)

```
message Buff {
uint32 robot_id = 1;
uint32 buff_type = 2;
int32 buff_level = 3;
uint32 buff_max_time = 4;
uint32 buff_left_time = 5;
string msg_params = 6;
}
```

## 2.2.14 PenaltyInfo

Purpose: Penalty information synchronization

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Current penalty type |
| 2 | uint32 | Current penalty duration (seconds) |
| 3 | uint32 | Current penalty count |

Enumeration of penalty_type values:

- 1: Yellow Card
- 2: Dual Yellow Card
- 3: Red Card
- 4: Overpower
- 5: Overheating
- 6: Exceeding firing rate

```
message PenaltyInfo {
uint32 penalty_type = 1;
uint32 penalty_effect_sec = 2;
uint32 total_penalty_num = 3;
}
```

## 2.2.15 RobotPathPlanInfo

Purpose: Sentry trajectory planning information

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Sentry intention (1 = Attack, 2 = Defense, 3 = Move) |
| 2 | uint32 | Starting point X coordinate (dm) |
| 3 | uint32 | Starting point Y coordinate (dm) |
| 4 | repeated int32 | Array of relative X increments from the starting point (-128 to +127, length 49) |
| 5 | repeated int32 | Array of relative Y-axis increments from the starting point (-128 to +127, length 49) |
| 6 | uint32 | Sender ID |

```
message RobotPathPlanInfo {
uint32 intention = 1;
uint32 start_pos_x = 2;
uint32 start_pos_y = 3;
repeated int32 offset_x = 4 [packed = true];
repeated int32 offset_y = 5 [packed = true];
uint32 sender_id = 6;
}
```

## 2.2.16 MapClickInfoNotify

Purpose: Gimbal Operator marking by clicking on the map

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Transmission range (0 = specified client, 1 = excluding Sentry, 2 = including Sentry) |
| 2 | bytes | List of target robots IDs (fixed at 7 bytes; when sending to Own Side Hero and Engineer, Red Team should enter 1, 2, and Blue Team should enter 101, 102, with the remaining digits filled with 0) |
| 3 | uint32 | Marker type (1 = Attack, 2 = Defense, 3 = Alert, 4 = Custom) |
| 4 | uint32 | ID of the marked opponent robot |
| 5 | uint32 | Custom icon ASCII code |
| 6 | uint32 | Marking mode (1 = Map, 2 = Opponent robots |
| 7 | uint32 | Screen X coordinate (pixels) |
| 8 | uint32 | Screen Y coordinate (pixels) |
| 9 | float | Map X coordinate |
| 10 | float | Map Y coordinate |

```
message MapClickInfoNotify {
uint32 is_send_all = 1;
bytes robot_id = 2;
uint32 mode = 3;
uint32 enemy_id = 4;
uint32 ascii = 5;
uint32 type = 6;
uint32 screen_x = 7;
uint32 screen_y = 8;
float map_x = 9;
float map_y = 10;
}
```

## 2.2.17 RadarInfoToClient

Purpose: Robot location information transmitted by Radar

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Target Robot ID |
| 2 | float | Target location X (in meters) |
| 3 | float | Target location Y (in meters) |

| Data ID | Data type | Data purpose |
|---|---|---|
| 4 | float | Orientation angle |
| 5 | uint32 | Is it specially marked (0 = No, 1 = Yes) |

```
message RaderInfoToClient {
uint32 target_robot_id = 1;
float target_pos_x = 2;
float target_pos_y = 3;
float orientation_angle = 4;
uint32 is_high_light = 5;
}
```

## 2.2.18 CustomByteBlock

Purpose: Custom data stream

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | bytes | Custom data packet up to 1.2 kbit |

```
message CustomByteBlock {
bytes data = 1;
}
```

## 2.2.19 AssemblyCommand

Purpose: Engineer assembly command

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Assembly operation type (1 = confirm assembly, 2 = cancel assembly) |
| 2 | uint32 | Selected assembly difficulty |

```
message AssemblyCommand {
uint32 operation = 1;
uint32 difficulty = 2;
}
```

## 2.2.20 TechCoreMotionStateSync

Purpose: Motion status synchronization for the Tech Core

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | The highest difficulty level currently available for assembly selection |
| 2 | uint32 | Tech Core Status |

Enumeration of status values:

- 1: Not yet in assembly mode
- 2: Difficulty level selected, Tech Core in motion
- 3: Tech Core movement complete, ready to begin the first assembly step
- 4: Previous assembly step completed, ready to proceed to the next assembly step.
- 5: All assembly steps are completed
- 6: Assembly confirmed, Tech Core in transit

```
message TechCoreMotionStateSync {
uint32 maximum_difficulty_level = 1;
uint32 status = 2;
}
```

## 2.2.21 RobotPerformanceSelectionCommand

Purpose: Infantry/Hero performance system selection

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Launching Mechanism Performance System |
| 2 | uint32 | Chassis Performance System |

```
message RobotPerformanceSelectionCommand {
uint32 shooter = 1;
uint32 chassis = 2;
}
```

## 2.2.22 RobotPerformanceSelectionSync

Purpose: Synchronize performance system status for Infantry/Hero

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Launching Mechanism Performance System |
| 2 | uint32 | Chassis Performance System |

```
message RobotPerformanceSelectionSync {
uint32 shooter = 1;
uint32 chassis = 2;
```

```
}
```

## 2.2.23 HeroDeployModeEventCommand

Purpose: Hero Deploy Mode command

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Mode (0 = Exit, 1 = Enter) |

```
message HeroDeployModeEventCommand {
uint32 mode = 1;
}
```

## 2.2.24 DeployModeStatusSync

Purpose: Hero Deployment Mode status synchronization

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Current deployment mode status (0 = not deployed, 1 = deployed) |

```
message DeployModeStatusSync {
uint32 status = 1;
}
```

## 2.2.25 RuneActivateCommand

Purpose: Power Rune activation command

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Activate (1 = On) |

```
message RuneActivateCommand {
uint32 activate = 1;
}
```

## 2.2.26 RuneStatusSync

Purpose: Power Rune status synchronization

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Enumeration of current Power Rune statuses |

| Data ID | Data type | Data purpose |
|---|---|---|
| 2 | uint32 | Current number of activated light arms |
| 3 | uint32 | Total number of rings |

Enumeration of rune_status values:

- 1: Not Activated
- 2: Activating
- 3: Activated

```
message RuneStatusSync {
uint32 rune_status = 1;
uint32 activated_arms = 2;
uint32 average_rings = 3;
}
```

## 2.2.27 SentinelStatusSync

Purpose: Sentry mode and weakened state

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Mode ID (1 for offensive, 2 for defensive , 3 for mobile ) |
| 2 | bool | Is weakened |

```
message SentinelStatusSync {
uint32 posture_id = 1;
bool is_weakened = 2;
}
```

## 2.2.28 DartCommand

Purpose: Dart control command

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Target ID (1: Outpost, 2: Base fixed point, 3: Base randomly assigned fixed point, 4: Base randomly assigned moving point, 5: Base terminal moving point) |
| 2 | bool | Gate switch |

```
message DartCommand {
uint32 target_id = 1;
bool open = 2;
}
```

## 2.2.29 DartSelectControlStatusSync

Purpose: Dart target selection status synchronization

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Target ID |
| 2 | bool | Gate status |

```
message DartSelectControlStatusSync {
uint32 target_id = 1;
bool open = 2;
}
```

## 2.2.30 GuardCtrlCommand

Purpose: Sentry Control command request

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Command ID (0 = invalid) |

Enumeration of command code :

- 1: Reload at Restoration Zone
- 2: Physical projectile reload at the supplier
- 3: Remote reload Remote reload
- 4: Remote HP recovery
- 5: Confirm respawn
- 6: Confirm to spend Gold Coin for respawn
- 7: Map marker Map marker
- 8: Switch to offensive mode
- 9: Switch to defensive mode Switch to defensive mode
- 10: Switch to mobile mode

```
message GuardCtrlCommand {
uint32 command_id = 1;
}
```

## 2.2.31 GuardCtrlResult

Purpose: Feedback on Sentry control command

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Corresponding Command ID |
| 2 | uint32 | Result code |

Enumeration of result_code values:

- 0: Success
- Miscellaneous: Failed

```
message GuardCtrlResult {
uint32 command_id = 1;
uint32 result_code = 2;
}
```

## 2.2.32 AirSupportCommand

Purpose: Air support command

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Command Type |

Enumeration of command_id values:

- 1: Request air support for free
- 2: Request air support using Gold Coin (free duration will still be used first)
- 3: Interrupt air support

```
message AirSupportCommand {
uint32 command_id = 1;
}
```

## 2.2.33 AirSupportStatusSync

Purpose: Air support status feedback

| Data ID | Data type | Data purpose |
|---|---|---|
| 1 | uint32 | Air Support Status |
| 2 | uint32 | Remaining time for free air support |
| 3 | uint32 | Spent Gold Coins for paid air support |

Enumeration of AirSupport_Status values:

- 0: Air support not activated
- 1: Air support in progress
- 2: Air support is locked by the opponent and cannot be activated

```
message AirSupportStatusSync {
uint32 airsupport_status = 1;
```

```
uint32 left_time = 2;
uint32 cost_gold_coin = 3;
}
```

# Appendix I: CRC check code example

```
//crc8 generator polynomial:G(x)=x8+x5+x4+1
const unsigned char CRC8_INIT = 0xff;
const unsigned char CRC8_TAB[256] =
{
0x00, 0x5e, 0xbc, 0xe2, 0x61, 0x3f, 0xdd, 0x83, 0xc2, 0x9c, 0x7e, 0x20, 0xa3, 0xfd, 0x1f, 0x41,
0x9d, 0xc3, 0x21, 0x7f, 0xfc, 0xa2, 0x40, 0x1e, 0x5f, 0x01, 0xe3, 0xbd, 0x3e, 0x60, 0x82, 0xdc, 0x23, 0x7d, 0x9f,
0xc1, 0x42, 0x1c, 0xfe, 0xa0, 0xe1, 0xbf, 0x5d, 0x03, 0x80, 0xde, 0x3c, 0x62, 0xbe, 0xe0, 0x02, 0x5c, 0xdf, 0x81,
0x63, 0x3d, 0x7c, 0x22, 0xc0, 0x9e, 0x1d, 0x43, 0xa1, 0xff, 0x46, 0x18, 0xfa, 0xa4, 0x27, 0x79, 0x9b, 0xc5, 0x84,
0xda, 0x38, 0x66, 0xe5, 0xbb, 0x59, 0x07, 0xdb, 0x85, 0x67, 0x39, 0xba, 0xe4, 0x06, 0x58, 0x19, 0x47, 0xa5,
0xfb, 0x78, 0x26, 0xc4, 0x9a, 0x65, 0x3b, 0xd9, 0x87, 0x04, 0x5a, 0xb8, 0xe6, 0xa7, 0xf9, 0x1b, 0x45, 0xc6,
0x98, 0x7a, 0x24, 0xf8, 0xa6, 0x44, 0x1a, 0x99, 0xc7, 0x25, 0x7b, 0x3a, 0x64, 0x86, 0xd8, 0x5b, 0x05, 0xe7,
0xb9,
0x8c, 0xd2, 0x30, 0x6e, 0xed, 0xb3, 0x51, 0x0f, 0x4e, 0x10, 0xf2, 0xac, 0x2f, 0x71, 0x93, 0xcd, 0x11, 0x4f, 0xad,
0xf3, 0x70, 0x2e, 0xcc, 0x92, 0xd3, 0x8d, 0x6f, 0x31, 0xb2, 0xec, 0x0e, 0x50, 0xaf, 0xf1, 0x13, 0x4d, 0xce, 0x90,
0x72, 0x2c, 0x6d, 0x33, 0xd1, 0x8f, 0x0c, 0x52, 0xb0, 0xee, 0x32, 0x6c, 0x8e, 0xd0, 0x53, 0x0d, 0xef, 0xb1, 0xf0,
0xae, 0x4c, 0x12, 0x91, 0xcf, 0x2d, 0x73, 0xca, 0x94, 0x76, 0x28, 0xab, 0xf5, 0x17, 0x49, 0x08, 0x56, 0xb4, 0xea,
0x69, 0x37, 0xd5, 0x8b, 0x57, 0x09, 0xeb, 0xb5, 0x36, 0x68, 0x8a, 0xd4, 0x95, 0xcb, 0x29, 0x77, 0xf4, 0xaa,
0x48, 0x16, 0xe9, 0xb7, 0x55, 0x0b, 0x88, 0xd6, 0x34, 0x6a, 0x2b, 0x75, 0x97, 0xc9, 0x4a, 0x14, 0xf6, 0xa8,
0x74, 0x2a, 0xc8, 0x96, 0x15, 0x4b, 0xa9, 0xf7, 0xb6, 0xe8, 0x0a, 0x54, 0xd7, 0x89, 0x6b, 0x35,
};
unsigned char Get_CRC8_Check_Sum(unsigned char *pchMessage,unsigned int dwLength,unsigned char ucCRC8)
{
unsigned char ucIndex;
while (dwLength--)
{
ucIndex = ucCRC8^(*pchMessage++);
ucCRC8 = CRC8_TAB[ucIndex];
}
return(ucCRC8);
}
/*
** Descriptions: CRC8 Verify function
** Input: Data to Verify,Stream length = Data + checksum
** Output: True or False (CRC Verify Result)
*/
unsigned int Verify_CRC8_Check_Sum(unsigned char *pchMessage, unsigned int dwLength)
{
unsigned char ucExpected = 0;
if ((pchMessage == 0) || (dwLength <= 2)) return 0;
ucExpected = Get_CRC8_Check_Sum (pchMessage, dwLength-1, CRC8_INIT);
return ( ucExpected == pchMessage[dwLength-1] );
}
```

```
/*
** Descriptions: append CRC8 to the end of data
** Input: Data to CRC and append,Stream length = Data + checksum
** Output: True or False (CRC Verify Result)
*/
void Append_CRC8_Check_Sum(unsigned char *pchMessage, unsigned int dwLength)
{
unsigned char ucCRC = 0;
if ((pchMessage == 0) || (dwLength <= 2)) return;
ucCRC = Get_CRC8_Check_Sum ( (unsigned char *)pchMessage, dwLength-1, CRC8_INIT);
pchMessage[dwLength-1] = ucCRC;
}

uint16_t CRC_INIT = 0xffff;
const uint16_t wCRC_Table[256] =
{
0x0000, 0x1189, 0x2312, 0x329b, 0x4624, 0x57ad, 0x6536, 0x74bf,
0x8c48, 0x9dc1, 0xaf5a, 0xbed3, 0xca6c, 0xdbe5, 0xe97e, 0xf8f7,
0x1081, 0x0108, 0x3393, 0x221a, 0x56a5, 0x472c, 0x75b7, 0x643e,
0x9cc9, 0x8d40, 0xbfdb, 0xae52, 0xdaed, 0xcb64, 0xf9ff, 0xe876,
0x2102, 0x308b, 0x0210, 0x1399, 0x6726, 0x76af, 0x4434, 0x55bd,
0xad4a, 0xbcc3, 0x8e58, 0x9fd1, 0xeb6e, 0xfae7, 0xc87c, 0xd9f5,
0x3183, 0x200a, 0x1291, 0x0318, 0x77a7, 0x662e, 0x54b5, 0x453c,
0xbdcb, 0xac42, 0x9ed9, 0x8f50, 0xfbef, 0xea66, 0xd8fd, 0xc974,
0x4204, 0x538d, 0x6116, 0x709f, 0x0420, 0x15a9, 0x2732, 0x36bb,
0xce4c, 0xdfc5, 0xed5e, 0xfcd7, 0x8868, 0x99e1, 0xab7a, 0xbaf3,
0x5285, 0x430c, 0x7197, 0x601e, 0x14a1, 0x0528, 0x37b3, 0x263a,
0xdecd, 0xcf44, 0xfddf, 0xec56, 0x98e9, 0x8960, 0xbbfb, 0xaa72,
0x6306, 0x728f, 0x4014, 0x519d, 0x2522, 0x34ab, 0x0630, 0x17b9,
0xef4e, 0xfec7, 0xcc5c, 0xddd5, 0xa96a, 0xb8e3, 0x8a78, 0x9bf1,
0x7387, 0x620e, 0x5095, 0x411c, 0x35a3, 0x242a, 0x16b1, 0x0738,
0xffcf, 0xee46, 0xdcdd, 0xcd54, 0xb9eb, 0xa862, 0x9af9, 0x8b70,
0x8408, 0x9581, 0xa71a, 0xb693, 0xc22c, 0xd3a5, 0xe13e, 0xf0b7,
0x0840, 0x19c9, 0x2b52, 0x3adb, 0x4e64, 0x5fed, 0x6d76, 0x7cff,
0x9489, 0x8500, 0xb79b, 0xa612, 0xd2ad, 0xc324, 0xf1bf, 0xe036,
0x18c1, 0x0948, 0x3bd3, 0x2a5a, 0x5ee5, 0x4f6c, 0x7df7, 0x6c7e,
0xa50a, 0xb483, 0x8618, 0x9791, 0xe32e, 0xf2a7, 0xc03c, 0xd1b5,
0x2942, 0x38cb, 0x0a50, 0x1bd9, 0x6f66, 0x7eef, 0x4c74, 0x5dfd,
0xb58b, 0xa402, 0x9699, 0x8710, 0xf3af, 0xe226, 0xd0bd, 0xc134,
0x39c3, 0x284a, 0x1ad1, 0x0b58, 0x7fe7, 0x6e6e, 0x5cf5, 0x4d7c,
0xc60c, 0xd785, 0xe51e, 0xf497, 0x8028, 0x91a1, 0xa33a, 0xb2b3,
```

```
0x4a44, 0x5bcd, 0x6956, 0x78df, 0x0c60, 0x1de9, 0x2f72, 0x3efb,
0xd68d, 0xc704, 0xf59f, 0xe416, 0x90a9, 0x8120, 0xb3bb, 0xa232,
0x5ac5, 0x4b4c, 0x79d7, 0x685e, 0x1ce1, 0x0d68, 0x3ff3, 0x2e7a,
0xe70e, 0xf687, 0xc41c, 0xd595, 0xa12a, 0xb0a3, 0x8238, 0x93b1,
0x6b46, 0x7acf, 0x4854, 0x59dd, 0x2d62, 0x3ceb, 0x0e70, 0x1ff9,
0xf78f, 0xe606, 0xd49d, 0xc514, 0xb1ab, 0xa022, 0x92b9, 0x8330,
0x7bc7, 0x6a4e, 0x58d5, 0x495c, 0x3de3, 0x2c6a, 0x1ef1, 0x0f78
};
/*
** Descriptions: CRC16 checksum function
** Input: Data to check,Stream length, initialized checksum
** Output: CRC checksum
*/
uint16_t Get_CRC16_Check_Sum(uint8_t *pchMessage,uint32_t dwLength,uint16_t wCRC)
{
Uint8_t chData;
if (pchMessage == NULL)
{
return 0xFFFF;
}
while(dwLength--)
{
chData = *pchMessage++;
(wCRC) = ((uint16_t)(wCRC) >> 8) ^ wCRC_Table[((uint16_t)(wCRC) ^ (uint16_t)(chData)) & 0x00ff];
}
return wCRC;
}

/*
** Descriptions: CRC16 Verify function
** Input: Data to Verify,Stream length = Data + checksum
** Output: True or False (CRC Verify Result)
*/
uint32_t Verify_CRC16_Check_Sum(uint8_t *pchMessage, uint32_t dwLength)
{
uint16_t wExpected = 0;
if ((pchMessage == NULL) || (dwLength <= 2))
{
return __FALSE;
}
```

```
wExpected = Get_CRC16_Check_Sum ( pchMessage, dwLength - 2, CRC_INIT);
return ((wExpected & 0xff) == pchMessage[dwLength - 2] && ((wExpected >> 8) & 0xff) == pchMessage[dwLength - 1]);
}

/*
** Descriptions: append CRC16 to the end of data
** Input: Data to CRC and append,Stream length = Data + checksum
** Output: True or False (CRC Verify Result)
*/
void Append_CRC16_Check_Sum(uint8_t * pchMessage,uint32_t dwLength)
{
uint16_t wCRC = 0;
if ((pchMessage == NULL) || (dwLength <= 2))
{
return;
}
wCRC = Get_CRC16_Check_Sum ( (U8 *)pchMessage, dwLength-2, CRC_INIT );
pchMessage[dwLength-2] = (U8)(wCRC & 0x00ff);
pchMessage[dwLength-1] = (U8)((wCRC >> 8)& 0x00ff);
```

# Appendix II: Instruction for ID numbers

Robot ID numbers are as follows:

- 1: Red Team Hero
- 2: Red Team Engineer
- 3/4/5: Red Team Infantry robots (corresponding to robot IDs 3–5
- 6: Red Team Drone
- 7: Red Team Sentry
- 8: Red Team Dart
- 9: Red Team Radar
- 10: Red Team Outpost
- 11: Red Team Base
- 101: Blue Team Hero
- 102: Blue Team Engineer
- 103/104/105: Blue Team Infantry robots (corresponding to robot IDs 3–5
- 106: Blue Team Drone
- 107: Blue Team Sentry
- 108: Blue Team Dart
- 109: Blue Team Radar
- 110: Blue Team Outpost
- 111: Blue Team Base

Player's Client ID is as follows:

- 0x0101: Red Team Hero Player's Client
- 0x0102: Red Team Engineer Player's Client
- 0x0103/0x0104/0x0105: Red Team Infantry Player's Client (corresponding to Robots ID 3–5)
- 0x0106: Red Team Drone Player's Client
- 0x016A: Blue Team Drone Player's Client
- 0x0165: Blue Team Hero Player's Client
- 0x0166: Blue Team Engineer Player's Client

- 0x0167/0x0168/0x0169: Blue Team Infantry Player’s Client (corresponding to Robots with ID 3 to 5)
- 0x8080: Referee system Server (used for sentry and Radar Decision Commands)

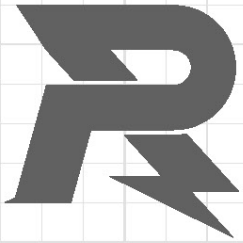
# Appendix III: Sample communication code for custom client

```
using MQTTnet;
using MQTTnet.Client;
using MQTTnet.Diagnostics;
using MQTTnet.Protocol;
using MQTTnet.Server;
using System;
using System.Collections;
using System.Collections.Generic;
using System.Text;
using System.Threading.Tasks;
using UnityEngine;

/// <summary>
/// MQTT Client
/// </summary>
public class MyMqttClient
{
    private IMqttClient m_MqttClient = null;
    private string m_ClientID;

    /// <summary>
    /// Create a client and connect to the server
    /// </summary>
    /// <param name="clientID"></param>
    /// <param name="ip"></param>
    /// <param name="port"></param>
    public MyMqttClient(string clientID, string ip = "127.0.0.1", int port = 3333)
    {
        m_ClientID = clientID;
        // Client options builder
        var options = new MqttClientOptionsBuilder()
        .WithClientId(m_ClientID)
        .WithTcpServer(ip, port)
        .Build();
        // Create client
        m_MqttClient = new MqttFactory().CreateMqttClient();
        // Monitor client connect or disconnect complete
        m_MqttClient.ConnectedAsync += ClientConnected;
        m_MqttClient.DisconnectedAsync += ClientDisConnected;
        // Client received message
        m_MqttClient.ApplicationMessageReceivedAsync += ReceiveMsg;
        // Connect to the server
        m_MqttClient.ConnectAsync(options);
    }

    /// <summary>
    /// Message received
    /// </summary>
    /// <param name="args"></param>
    /// <returns></returns>
    private Task ReceiveMsg(MqttApplicationMessageReceivedEventArgs args)
    {
     Debuger.Log(DebugLogEnum.CustomClient, $"recv topic: {args.ApplicationMessage.Topic}");
```

```
        //Message post appears to avoid message blockage
        MqttSubProtoEvent ev = new MqttSubProtoEvent();
        ev.topic = args.ApplicationMessage.Topic;
        ev.data = args.ApplicationMessage.Payload;
        ThreadPool.CustomClientLoop.PostEvent(ev);

        return Task.CompletedTask;
    }

    /// <summary>
    /// Disconnection completed
    /// </summary>
    /// <param name="args"></param>
    /// <returns></returns>
    private Task ClientDisConnected(MqttClientDisconnectedEventArgs args)
    {
        return Task.CompletedTask;
    }

    /// <summary>
    /// Connection completed
    /// </summary>
    /// <param name="args"></param>
    /// <returns></returns>ReceiveMsg
    private Task ClientConnected(MqttClientConnectedEventArgs args)
    {
       Subscribe("RemoteControl");
        Subscribe("CustomRobotData");

        return Task.CompletedTask;
    }

    /// <summary>
    /// Message release
    /// </summary>
       public void PublishMsg(string topic, string message, MqttQualityOfServiceLevel level =
MqttQualityOfServiceLevel.ExactlyOnce, bool isRetain = false)
    {
        long unixUs = DateTime.UtcNow.Ticks / 10; // microsecond-level
        m_MqttClient.PublishStringAsync(topic, message, level, isRetain);
    }
    /// <summary>
    /// Message release
    /// </summary>
      public void PublishBytesMsg(string topic, byte[] message, MqttQualityOfServiceLevel level =
MqttQualityOfServiceLevel.ExactlyOnce, bool isRetain = false)
    {
        long unixUs = DateTime.UtcNow.Ticks / 10; // microsecond-level
        m_MqttClient.PublishBinaryAsync(topic, message, level, isRetain);
    }

    /// <summary>
    /// Subscribe topic
    /// </summary>
    public void Subscribe(string topic)
    {
        m_MqttClient.SubscribeAsync(new MqttTopicFilterBuilder().WithTopic(topic).Build());
    }
}
```



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